Twistor–SMRK Framework for Yang–Mills Mass Gap

A QFC Operator Program on Symplectic Spinor Bundles with a Testable $\lambda \to 0$ Criterion

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Abstract

The Yang–Mills existence and mass gap problem asks for a mathematically rigorous construction of four-dimensional quantum Yang–Mills theory and a proof that the associated Hamiltonian has a strictly positive spectral gap above the vacuum. This whitepaper proposes a QFC operator program that couples two ingredients: (i) the symplectic twistor-operator viewpoint from symplectic spinor geometry ($\mathrm{Mp}(2n, \mathbb{R})$-structured state spaces), and (ii) a SMRK-inspired multi-resolution confinement regularization used to control low-energy leakage in operator spectra.

We construct a regularized family of Hamiltonians $\widetilde{H}_\lambda$ and formulate a precise $\lambda \to 0$ *criterion* under which spectral-gap bounds can persist in the physical limit. The program is deliberately written as a separation of what is proven, what is conditional, and what is falsifiable: the central quantitative test is whether $\Delta(\lambda)$ remains uniformly bounded away from zero or collapses as $\Delta(\lambda) \sim \lambda$.

Beyond the mass-gap target, the framework yields a structured decomposition of state space based on metaplectic symmetry and analytic spinor modes, providing a controlled interface to lattice validation and toy truncation experiments.

1 Introduction and Scope

1.1 The Clay Problem and the Spectral Target

The Yang–Mills existence and mass gap problem asks for a rigorous construction of four-dimensional quantum Yang–Mills theory for any compact simple gauge group G and a proof that the corresponding Hamiltonian admits a strictly positive spectral gap above the vacuum.

In operator language, the mass gap is the statement that if E_0 denotes the vacuum energy, then

$$\Delta := \inf\{E \in \mathrm{spec}(H) : E > E_0\} - E_0 > 0.$$

This is not a perturbative claim: it concerns the global structure of the spectrum.

1.2 What This Whitepaper Claims

This document introduces a QFC operator program that couples: (i) a symplectic-spinor state space carrying metaplectic symmetry, and (ii) a SMRK-inspired multi-resolution confinement term used as a regulator.

The central object is a family of regularized Hamiltonians

$$\widetilde{H}_\lambda := H_{\mathrm{YM}} + \lambda\, C_\lambda, \qquad \lambda > 0,$$

together with an explicit criterion under which spectral-gap bounds may persist as $\lambda \to 0$.

1.3 What This Whitepaper Does Not Claim

We do not claim, in this draft, a completed Clay-proof of existence and mass gap. In particular, the implication

$$\Delta(\widetilde{H}_\lambda) > 0 \;\;\not\Rightarrow\;\; \Delta(H_{\mathrm{YM}}) > 0$$

is treated as a nontrivial gate whose resolution requires an explicit $\lambda \to 0$ control statement (e.g. strong resolvent convergence together with uniform gap bounds).

1.4 Why Twistor–SMRK?

The symplectic twistor operator provides a structured analytic and representation- theoretic decomposition of the relevant state space, while the SMRK-inspired term acts as a spectral regularizer that can be measured and stress-tested numerically. The intended outcome is not a metaphor but a falsifiable operator program: the key test is whether $\Delta(\lambda)$ remains uniformly bounded away from zero or collapses with λ.

2 Canonical Yang–Mills Hamiltonian and the Real Obstruction

2.1 Hamiltonian Formulation and the Physical Hilbert Space

We work in the canonical (Hamiltonian) formulation of Yang–Mills theory with a compact simple gauge group G (e.g. $SU(N)$). On a spatial slice $\mathbb{R}^3$, the classical phase space is coordinatized by a gauge potential $A_i(x)$ and its conjugate electric field $E^i(x)$, valued in the Lie algebra $\mathfrak{g}$. Formally, the phase space carries the usual symplectic (Poisson) structure, and quantization promotes (A, E) to operator-valued distributions.

Gauge invariance imposes a first-class constraint (Gauss law). In operator form, physical states must satisfy

$$\widehat{D}_i \widehat{E}^{\,i}\, |\psi\rangle = 0,$$

where D_i is the gauge-covariant derivative. The physical Hilbert space is therefore not a naive Fock space of free gluons, but a gauge-invariant subspace (or quotient construction) in which Gauss-law constraints are implemented at the operator level. This is already a structural distinction from perturbative treatments where gauge fixing and BRST are used as computational devices.

2.2 Formal Hamiltonian and its Decomposition

In Coulomb-type gauges (or any complete gauge fixing where the constraint is explicitly handled), the formal Yang–Mills Hamiltonian takes the form

$$\widehat{H}_{\mathrm{YM}} = \frac{g^2}{2} \int_{\mathbb{R}^3} \left(\widehat{E}_i^{\,a} \widehat{E}_i^{\,a} + \widehat{B}_i^{\,a} \widehat{B}_i^{\,a} \right) d^3x \;+\; \widehat{H}_{\mathrm{Coulomb}},$$

where a indexes a basis of $\mathfrak{g}$, g is the coupling constant, and

$$B_i^a = \frac{1}{2}\,\varepsilon_{ijk}\,F_{jk}^a, \qquad F_{jk}^a = \partial_j A_k^a - \partial_k A_j^a + f^{abc} A_j^b A_k^c.$$

The term $\widehat{H}_{\text{Coulomb}}$ is a non-local functional induced by solving the Gauss constraint in a gauge-fixed setting, and it is conceptually important because it is not merely a perturbative correction: it is part of the true Hamiltonian content once gauge invariance is enforced.

For the purposes of this program, it is useful to separate the Hamiltonian into

$$\widehat{H}_{\text{YM}} = \widehat{H}_{\text{free}} + \widehat{H}_{\text{int}} \quad \text{(plus Coulomb/nonlocal terms as required by the gauge choice)},$$

where $\widehat{H}_{\text{free}}$ is the quadratic (free-field) part, and $\widehat{H}_{\text{int}}$ collects the cubic and quartic self-interactions arising from $F = dA + A \wedge A$.

2.3 Why the Free Theory Has No Gap

The free (massless) field Hamiltonian is, morally, an infinite collection of harmonic oscillators with dispersion $\omega(k) = |k|$. Formally, its spectrum is continuous from 0 upward:

$$\mathrm{spec}(\widehat{H}_{\text{free}}) = \{0\} \cup [0, \infty), \qquad \Delta_{\text{free}} = 0.$$

This is not a defect but the expected behavior of a massless gauge field. Therefore, a mass gap in Yang–Mills is not inherited from the free theory; it is a *non-perturbative spectral phenomenon* produced by interaction, constraint, and the global structure of the physical state space.

2.4 The Real Obstruction: Operator Meaning Before Spectrum

At the Clay-problem level, the Yang–Mills challenge is often presented as a spectral-gap statement. However, the deeper order of difficulty is:

1. Construct an actual Hilbert space of physical states (respecting gauge constraints) on which the Hamiltonian acts.
2. Define $\widehat{H}_{\text{YM}}$ as a densely defined operator with a specified domain $\mathcal{D}(\widehat{H}_{\text{YM}})$.
3. Prove the existence of a self-adjoint realization (or essential self-adjointness) so that the spectrum is mathematically meaningful.
4. Only then: analyze $\mathrm{spec}(\widehat{H}_{\text{YM}})$ and prove a gap.

In other words, "mass gap" is downstream from "operator existence."

From a QFC perspective, this is an ideal problem archetype: the object of interest is explicitly a (candidate) self-adjoint Hamiltonian whose spectrum encodes a physical invariant. But it also clarifies why naive arguments fail: most "gap" intuitions implicitly assume a well-defined self-adjoint operator, while the Clay formulation requires constructing one.

2.5 The Regularization Trap: A Gap for $\widetilde{H}_\lambda$ is Not Yet a Gap for H

A standard maneuver—and the starting point of our program—is to introduce a controlled regularization that tames low-energy leakage or domain pathologies by adding a confining term:

$$\widetilde{H}_\lambda := H_{\text{YM}} + \lambda\, C_\lambda, \qquad \lambda > 0.$$

For suitable C_λ, it may be feasible to prove: (i) $\widetilde{H}_\lambda$ is self-adjoint for each fixed $\lambda > 0$, and (ii) $\Delta(\lambda) := \Delta(\widetilde{H}_\lambda) > 0$.

The *real obstruction* is that neither statement, by itself, implies the Clay target. The implication

$$\Delta(\widetilde{H}_\lambda) > 0 \quad \Rightarrow \quad \Delta(H_{\mathrm{YM}}) > 0$$

is false in general. A regulator can create an artificial gap that collapses as $\lambda \to 0$:

$$\Delta(\lambda) \sim \lambda \quad \Longrightarrow \quad \lim_{\lambda \to 0} \Delta(\lambda) = 0.$$

Therefore the central gate of the entire approach is the $\lambda \to 0$ problem:

> *Can one control the $\lambda \to 0$ limit in a way that preserves both the operator meaning and a strictly positive spectral threshold?*

2.6 What Must Be Proven in Any Successful Program

To convert a "regularized gap" into a "physical gap," one needs a pair of statements of the following type:

1. **Operator convergence:** $\widetilde{H}_\lambda \to H_{\mathrm{YM}}$ in a topology strong enough to control spectral behavior (typically, a resolvent-based notion such as strong resolvent convergence).
2. **Uniformity:** a lower bound $\Delta(\lambda) \geq \Delta_0 > 0$ that does not deteriorate as $\lambda \to 0$.

Only with both ingredients does it become plausible to transfer a gap from the regularized family to the limiting operator. This whitepaper is organized around making these gates explicit and designing a Twistor–SMRK structure in which they can be attacked in a controlled, falsifiable manner.

2.7 QFC Framing of the Obstruction

In QFC terms, the obstruction is not "lack of physical intuition," but lack of a verified operator pipeline:

$$(\text{states}) \;\to\; (\text{self-adjoint operator}) \;\to\; (\text{spectral commitment}).$$

The Twistor–SMRK program is proposed precisely to supply a disciplined route through this pipeline: twistor structure is used to impose analytic and representation-theoretic control on state space, while SMRK-style multi-resolution confinement is used as an explicit regulator whose spectral effect can be measured, bounded, and stress-tested.

In the next section we state the QFC design principles that govern the operator choices in this document, including how "verification-first" thinking constrains what counts as an acceptable regularization and what counts as a valid $\lambda \to 0$ transfer theorem.

3 QFC Design Principles for Field Operators

3.1 Operator-First Construction vs. Formal Expansions

Quantum Field Computing (QFC) treats a quantum field theory as a *verified operator system* rather than a formal expansion scheme. In standard perturbative practice, one often proceeds by defining correlation functions through a formal path integral and extracting predictions via series expansions. While extraordinarily successful for many purposes, this workflow does not by itself satisfy the Clay requirement: the Millennium statement demands a mathematically well-defined quantum theory and a meaningful spectral claim about its Hamiltonian.

In QFC, the direction of construction is reversed:

$$\textbf{(state space)} \Rightarrow \textbf{(operator meaning)} \Rightarrow \textbf{(spectrum as an observable)}.$$

The Yang–Mills mass gap is then interpreted as a *structural spectral invariant* that must be produced by an explicitly defined self-adjoint operator, not merely suggested by asymptotic expansions.

3.2 Execution as a First-Class Mathematical Object

A core QFC stance is that "computation" and "physics" meet at the level of *execution artifacts*. In this setting, an operator is not only an abstract mapping but also an object that can be:

1. **Admitted:** executed only under a specified rule-set (domain, symmetry constraints, normalization rules).
2. **Traced:** associated with a reproducible record of how spectral or expectation-value claims were obtained (toy truncations, lattice discretizations, basis expansions, etc.).
3. **Compared:** evaluated by independent implementations that should agree on invariant outputs up to controlled error.

For Yang–Mills, this matters because the decisive question is not merely "does a gap exist?", but *what pipeline turns that question into a stable, auditable operator statement?*

3.3 Three QFC Operator Axioms (Minimal Version)

This whitepaper follows three minimal design axioms. They are intentionally conservative: they do not solve the problem by definition, but they constrain what we are allowed to call a valid step.

Axiom 1 (Explicit Domain). Every operator introduced must come with a stated dense domain. If an operator is given as a "formal expression" (e.g. involving fields as distributions), it is not considered part of the verified pipeline until a domain and closure/self-adjoint extension are specified.

Axiom 2 (Symmetry as a Constraint, Not a Slogan). Gauge invariance and the constraint structure are treated as operator-level requirements. "Working in a gauge" is acceptable only if the constraint is implemented consistently so that the resulting operator acts on a physical state space with a clear interpretation.

Axiom 3 (Spectral Claims Require Transfer Theorems). Any spectral claim derived for a regularized operator family (cutoffs, lattice, λ-terms, truncations) must be accompanied by a clearly stated transfer problem. In particular, a gap shown for $\widetilde{H}_\lambda$ is treated as *conditional evidence* unless a $\lambda \to 0$ (or cutoff removal) theorem is provided.

3.4 Why QFC Needs "Multi-Resolution" in Field Problems

A recurring obstacle in infinite-dimensional spectral problems is low-energy leakage: spectra that accumulate at 0, or sequences of states that escape to regions of configuration space where the intended operator control degrades. In QFM/SMRK-style constructions, multi-resolution structure is used to prevent such leakage by decomposing the state space into controlled channels and imposing a confinement mechanism that penalizes uncontrolled excursions.

For Yang–Mills, the analogous goal is to introduce a regulator that:

1. is explicit enough to be analyzed as an operator (not just a heuristic);
2. respects the relevant symmetry constraints (or breaks them only in a controlled and reversible way);
3. can be tested numerically to detect whether it generates a fake gap.

The Twistor–SMRK proposal is motivated precisely by this: the twistor backbone provides analytic and representation-theoretic control over modes, while the SMRK component supplies a tunable confinement term that is intended to stabilize the low-energy sector without destroying the possibility of removing the regulator.

3.5 What "Verification-First" Means for Regularization

Not all regularizations are acceptable in a QFC sense. We adopt the following verification-first criteria for any added term C_λ:

1. **Operator legibility:** C_λ should be a recognizable operator class (e.g. a multiplication operator on a function space, a form-bounded perturbation, or a closed quadratic form).
2. **Domain compatibility:** the domain $\mathcal{D}(H_{\mathrm{YM}})$ (or its candidate core) must interact with C_λ in a way that admits standard extension theorems (Kato–Rellich, form methods, etc.).
3. **Removability:** there must be a plausible path to $\lambda \to 0$ (or cutoff removal) with an explicit convergence target. Otherwise the construction is only a new theory, not a route to the Clay object.
4. **Falsifiability:** the spectral effect must be testable. The key falsification in this program is the scaling behavior of $\Delta(\lambda)$ as $\lambda \to 0$.

3.6 The Central QFC Test for the Yang–Mills Program

We emphasize a single quantitative decision criterion that will appear repeatedly:

> **Uniform-gap test.** Compute or bound $\Delta(\lambda)$ for a sequence of decreasing λ in validated truncations/discretizations. If $\Delta(\lambda)$ collapses as $\Delta(\lambda) \sim \lambda$ (or faster), the gap is likely a regulator artifact. If $\Delta(\lambda)$ stabilizes to a nonzero plateau, the program earns the right to pursue a $\lambda \to 0$ transfer theorem.

This test is not presented as a proof, but as the minimal verification-first gate that prevents the program from drifting into unfalsifiable narrative. It also provides a direct interface to the validation layer: finite-mode toys and lattice implementations can probe the scaling early.

3.7 Positioning of the Twistor Backbone

The twistor component is not introduced for aesthetic reasons; it is introduced because it offers two QFC-relevant capabilities:

1. **Analytic regularity control:** twistor equations impose PDE constraints whose solution spaces have strong regularity properties.
2. **Representation-structured mode decomposition:** metaplectic symmetry organizes the state space into invariant subspaces, allowing the operator program to be expressed channel-by-channel rather than as a single opaque infinite-dimensional object.

In the next sections we formalize the symplectic-spinor setting and the specific symplectic twistor operator used in this program, and we state precisely which parts are imported as definitions, which parts are proven, and which parts remain conditional.

4 Symplectic Linear Algebra and Metaplectic Symmetry

4.1 Symplectic Vector Spaces

Let V be a real vector space of even dimension $2n$. A *symplectic form* on V is a bilinear form $\Omega : V \times V \to \mathbb{R}$ that is *antisymmetric* and *non-degenerate*:

$$\Omega(v, w) = -\Omega(w, v), \qquad (\Omega(v, u) = 0 \ \forall u \in V) \Rightarrow v = 0.$$

A pair (V, Ω) is called a *symplectic vector space*. A standard result states that (V, Ω) admits a *symplectic basis* $\{e_1, \dots, e_n, e_{n+1}, \dots, e_{2n}\}$ such that

$$\Omega(e_i, e_j) = 0 = \Omega(e_{n+i}, e_{n+j}), \qquad \Omega(e_i, e_{n+j}) = \delta_{ij},$$

and, in matrix form,

$$\Omega(v, u) = -v^{\mathsf{T}} J\, u, \qquad J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

4.2 The Symplectic Group $\mathrm{Sp}(2n, \mathbb{R})$

The *symplectic group* is the group of linear automorphisms preserving Ω:

$$\mathrm{Sp}(V, \Omega) := \{A \in \mathrm{GL}(V) \ : \ \Omega(Av, Aw) = \Omega(v, w) \ \forall v, w \in V\}.$$

In the standard matrix realization with J as above,

$$\mathrm{Sp}(2n, \mathbb{R}) = \{A \in \mathrm{GL}(2n, \mathbb{R}) \ : \ A^{\mathsf{T}} J A = J\}.$$

This group controls the linear symmetries of the canonical phase space of a Hamiltonian system, and it is the natural symmetry group behind the "phase-space quantization" viewpoint.

4.3 Heisenberg Group and Canonical Quantization

Let (V, Ω) be symplectic. The associated *Heisenberg group* $H(V)$ is the set $V \times \mathbb{R}$ with multiplication

$$(v, t) \cdot (w, s) = \Big(v + w, \ t + s + \tfrac{1}{2}\Omega(v, w)\Big).$$

Its center is $\{(0, t) : t \in \mathbb{R}\}$. The role of $H(V)$ in quantization is that its (projective) unitary representations encode the canonical commutation relations.

In particular, in the Schrödinger picture one obtains unitary operators that implement translations and modulations on $L^2(\mathbb{R}^n)$, and the Stone–von Neumann theorem asserts essential uniqueness (up to unitary equivalence) of irreducible unitary representations with fixed central character.

4.4 The Metaplectic Group $\mathrm{Mp}(2n, \mathbb{R})$

The *metaplectic group* $\mathrm{Mp}(2n, \mathbb{R})$ is a double cover of $\mathrm{Sp}(2n, \mathbb{R})$:

$$1 \longrightarrow \{\pm 1\} \longrightarrow \mathrm{Mp}(2n, \mathbb{R}) \xrightarrow{\lambda} \mathrm{Sp}(2n, \mathbb{R}) \longrightarrow 1.$$

It arises because the symplectic group acts on the Heisenberg group by automorphisms, and lifting this action to the level of unitary operators on a Hilbert space generally produces only a *projective* representation. The metaplectic group is the minimal covering group on which the action becomes (unitarily) representable.

Concretely, one obtains a (projective) unitary representation U satisfying

$$U(ab) = c(a, b)\, U(a)\, U(b),$$

for a cocycle $c(a, b) \in S^1$, and passing to $\mathrm{Mp}(2n, \mathbb{R})$ eliminates the projective ambiguity.

4.5 Why Metaplectic Symmetry Matters for This Whitepaper

The Twistor–SMRK program requires a state space that is simultaneously:

1. compatible with canonical quantization (Heisenberg structure),
2. stable under a large symmetry group organizing "modes" into invariant channels, and
3. sufficiently regular to admit analytic control of operator domains.

Metaplectic symmetry is the natural symmetry of the Schrödinger/Segal–Shale–Weil quantization of symplectic phase space, and it will be used in two ways:

- to define and control the symplectic-spinor state space on which the twistor operator acts;
- to obtain invariant subspaces (in particular, kernels of invariant differential operators) that serve as structured "channels" for the subsequent SMRK-style multi-resolution analysis.

In the next section we introduce the analytic symplectic spinor space used in this paper and its canonical Hermite-mode decomposition, which provides the multi-resolution backbone for the SMRK coupling.

5 The Space of Symplectic Spinors

5.1 Overview and Role in the Program

The Twistor–SMRK framework requires a function space that simultaneously supports: (i) canonical quantization (Heisenberg structure), (ii) metaplectic symmetry ($\mathrm{Mp}(2n, \mathbb{R})$ action), and (iii) analytic regularity strong enough to make differential operators and mode decompositions well behaved.

Following the symplectic spinor approach, we work with an analytic "spinor" space denoted A, realized as a subspace of $L^2(\mathbb{R}^n)$ that is dense, stable under Fourier transform, and equipped with a Fréchet topology. This space serves as the local model for symplectic spinor bundles in the geometric part of the theory.

5.2 Fréchet Spaces and Why They Appear Here

A *Fréchet space* is a complete locally convex topological vector space whose topology can be defined by a countable family of seminorms (equivalently, by a translation-invariant metric inducing completeness). Fréchet spaces are natural in analysis of smooth and rapidly decreasing functions: they allow control by families of seminorms corresponding to derivatives and growth bounds.

We emphasize that working in a Fréchet setting is not a stylistic preference: it is a functional-analytic tool for handling spaces where Banach norms are too coarse to capture simultaneously (a) rapid decay, (b) analyticity, and (c) closure properties under operator actions.

5.3 The Analytic Rapid-Decay Space A

Let $A \subset L^2(\mathbb{R}^n)$ denote a space of rapidly decreasing analytic functions (a Fréchet space) that is dense in $L^2(\mathbb{R}^n)$ and behaves similarly to the Schwartz space $\mathcal{S}(\mathbb{R}^n)$, but with an explicit analytic structure.

For this whitepaper, we require the following operational features:

1. **Density:** A is dense in $L^2(\mathbb{R}^n)$.
2. **Stability:** A is stable under the Fourier transform and under the basic operations used in symplectic quantization (multiplication by polynomials, differentiation, etc.).
3. **Topological control:** convergence in A implies uniform control of analytic representatives on bounded sets (via seminorm estimates).

These properties are exactly what allow us to treat certain differential operators as continuous maps on appropriate spaces of A-valued smooth functions, and to speak meaningfully about kernels as closed invariant subspaces.

5.4 Fourier Transform as a Structural Symmetry

The Fourier transform is a central actor in phase-space quantization and in metaplectic representation theory. We assume the Fourier transform $\mathcal{F}$ acts bijectively on A and is continuous with respect to the Fréchet topology. This ensures that A supports both "position" and "momentum" descriptions of modes, and that switching between them does not leave the verified function space.

In QFC language: Fourier stability means that multi-resolution decompositions and spectral tests can be implemented consistently across dual representations of the same underlying state.

5.5 Hermite Functions and Canonical Mode Decomposition

A key reason to choose A is that it admits a canonical orthogonal mode decomposition based on Hermite functions. Let H_n denote the Hermite polynomials in one variable, defined by

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2},$$

and define the normalized Hermite functions

$$\psi_n(x) = \frac{1}{(2^n n! \sqrt{\pi})^{1/2}} \, e^{-x^2/2} \, H_n(x).$$

Then $\{\psi_n\}_{n\geq 0}$ forms an orthonormal basis of $L^2(\mathbb{R})$. In higher dimensions, tensor products yield an orthonormal basis $\{\psi_{\mathbf{n}}\}_{\mathbf{n}\in\mathbb{N}_0^n}$ of $L^2(\mathbb{R}^n)$.

Two recurrence relations will be repeatedly used in the SMRK coupling layer:

$$H_{n+1}(x) - 2xH_n(x) + 2nH_{n-1}(x) = 0, \qquad \frac{d}{dx}H_n(x) = 2nH_{n-1}(x).$$

Operationally, these identities provide explicit "ladder" couplings between adjacent resolution levels and are the mechanism by which SMRK-style interference can be expressed algebraically rather than heuristically.

5.6 Multi-Resolution Interpretation (QFC View)

In this whitepaper, we interpret Hermite modes as a *canonical multi-resolution coordinate system* on the analytic spinor space:

- low indices represent coarse, low-frequency structure;
- high indices represent fine, high-frequency structure;
- recurrence relations implement controlled couplings across scales.

This is not a wavelet system in the usual signal-processing sense, but it plays a similar organizational role: it provides an explicit decomposition in which operator actions can be tracked scale-by-scale.

5.7 Metaplectic Action and Invariant Subspaces

The metaplectic group acts (unitarily or projectively, depending on conventions) on suitable function spaces built from $L^2(\mathbb{R}^n)$. We assume the chosen spinor space A is stable under the (analytic) Segal–Shale–Weil type action. This matters because kernels of Mp-invariant operators become invariant subspaces, allowing the state space to be decomposed into representation-theoretic channels.

In the next section, we define the symplectic twistor operator acting on appropriate spaces of A-valued smooth functions, and we identify the invariance/regularity properties that make it suitable as a structural backbone for the Twistor–SMRK coupling.

6 The Symplectic Twistor Operator

6.1 Why a Twistor Operator Appears in This Program

The central role of the twistor operator in this whitepaper is structural: it provides a first-order analytic constraint whose solution space (kernel) is both regular and symmetry-organized. In the QFC framing, this supplies two critical capabilities:

1. **Analytic control:** the twistor equation enforces PDE regularity on admissible spinor fields, which is a direct lever on operator domains.
2. **Channel decomposition:** the kernel (and related subspaces) form invariant channels under metaplectic symmetry, which is the backbone needed for a clean multi-resolution (SMRK-style) coupling.

We emphasize again: twistor structure is not invoked here as Penrose twistor geometry on $\mathbb{CP}^3$, but as the *symplectic twistor operator* from symplectic spinor geometry. Any bridge to Penrose twistor constructions is postponed to future work to avoid mixing geometries.

6.2 Geometric Setting (Local Model)

Let (M, ω) be a symplectic manifold of dimension $2n$ equipped with a metaplectic structure. Let $\mathcal{Q} \to M$ denote the metaplectic principal bundle and let A be the symplectic spinor space introduced in Section 5. The associated symplectic spinor bundle is

$$\mathcal{S} := \mathcal{Q} \times_m A,$$

where m denotes the (Segal–Shale–Weil type) metaplectic representation on A.

A symplectic connection ∇ on (M, ω) induces a spinor covariant derivative ∇^S on $\mathcal{S}$, and hence an exterior spinor derivative

$$d_{\nabla^S} : \Omega^r(M, \mathcal{S}) \longrightarrow \Omega^{r+1}(M, \mathcal{S}).$$

Operationally, one may think of ∇^S as the device that allows us to take derivatives of spinor fields in a way compatible with the metaplectic structure.

6.3 Symplectic Clifford Multiplication (Operational Use)

The symplectic spinor bundle carries an action of tangent vectors by *symplectic Clifford multiplication* (sometimes called the symplectic spinor multiplication). Denote this action by

$$v \cdot \phi, \qquad v \in TM, \ \phi \in \Gamma(\mathcal{S}).$$

This action is the symplectic analogue of the orthogonal Clifford action used in Dirac-type constructions. In this paper we will use it only as an abstract operator that is:

- bilinear in v and ϕ,
- compatible with the metaplectic structure,
- suitable for defining first-order differential operators by contracting derivatives with Clifford multiplication.

6.4 Definition of the Symplectic Twistor Operator

The symplectic twistor operator is a first-order differential operator acting on spinor fields. There are several equivalent constructions; for the QFC program we adopt the following operational definition:

Definition 6.1 (Twistor operator). Let ∇^S be the symplectic spinor covariant derivative. Consider the map

$$\nabla^S : \Gamma(\mathcal{S}) \longrightarrow \Omega^1(M, \mathcal{S}), \qquad \phi \mapsto \nabla^S \phi.$$

Decompose $\Omega^1(M, \mathcal{S})$ into metaplectically natural components, and let $\mathcal{P}$ denote the projection onto the twistor component. Then the *symplectic twistor operator* is

$$T := \mathcal{P} \circ \nabla^S : \Gamma(\mathcal{S}) \longrightarrow \Gamma(\mathcal{T}),$$

where $\mathcal{T}$ denotes the target bundle (the "twistor" subbundle of $\Omega^1(M, \mathcal{S})$).

Remark. The concrete form of the projection $\mathcal{P}$ depends on the representation-theoretic decomposition of $T^*M \otimes \mathcal{S}$ under $\mathrm{Mp}(2n, \mathbb{R})$. The only features we require in this paper are: (i) T is first-order, (ii) T is Mp-equivariant, and (iii) $\ker(T)$ is a closed invariant subspace under the metaplectic action.

6.5 Local PDE Form and the Kernel Equation

In local coordinates (or in a local trivialization of the spinor bundle), the twistor equation $T\phi = 0$ becomes a system of linear first-order PDEs whose solutions are analytic and strongly constrained. In explicit models of symplectic twistor geometry, this kernel condition can be written as a PDE relating derivatives with respect to base coordinates and derivatives in spinor variables.

For the purposes of the QFC program, we adopt the guiding fact:

> **Kernel regularity principle.** The space $\ker(T)$ is highly regular (analytic) and is stable under the natural metaplectic symmetry; it therefore provides a disciplined subspace on which spectral constructions can be organized and tested channel-by-channel.

6.6 Metaplectic Equivariance and Invariant Channels

Because ∇^S is constructed from a symplectic connection and the bundle $\mathcal{S}$ is associated to the metaplectic structure, the twistor operator is $\mathrm{Mp}(2n, \mathbb{R})$-equivariant in the sense that the following diagram commutes (up to the representation conventions):

$$\begin{array}{ccc} \Gamma(\mathcal{S}) & \xrightarrow{\quad T \quad} & \Gamma(\mathcal{T}) \\ \big\downarrow m(g) & & \big\downarrow m_T(g) \\ \Gamma(\mathcal{S}) & \xrightarrow{\quad T \quad} & \Gamma(\mathcal{T}) \end{array} \qquad (g \in \mathrm{Mp}(2n, \mathbb{R})).$$

As a consequence, $\ker(T)$ is an invariant subspace:

$$\phi \in \ker(T) \implies m(g)\phi \in \ker(T).$$

This invariance is the technical reason the twistor backbone integrates cleanly with SMRK multi-resolution: it provides symmetry-stable channels on which multi-scale penalties can be applied without destroying the organizing symmetry.

6.7 How the Twistor Operator Enters the QFC Regularization

In this whitepaper, the twistor operator will not be used as a replacement for the Yang–Mills curvature F or for the gauge-covariant derivative D_A. Instead, it enters as a *state-space filter* used in the confinement term: the regularizer will penalize (in a controlled way) components of states that exhibit twistor-incoherent behavior.

Concretely, the confinement functional will involve quantities of the schematic form

$$\|T(\Phi)\|_{L^2}^2,$$

where Φ denotes the relevant spinor-valued object (a state, or a mode component of a state) in the chosen representation. The precise operator class of the resulting confinement term (multiplication operator vs. quadratic form) will be specified in Section 7.

6.8 Design Constraint: Do Not Mix Geometries

Finally, we state an explicit constraint that prevents conceptual drift:

> **No-geometry-mixing rule (v1).** All twistor constructions in this draft are symplectic (metaplectic) twistor constructions. Penrose twistor geometry on $\mathbb{CP}^3$ is treated, if at all, as a separate optional bridge in a dedicated future section, not as an input to the main operator proofs.

This rule ensures that the program remains mathematically legible and that every claimed invariance can be traced to the relevant symmetry group $\mathrm{Mp}(2n, \mathbb{R})$ rather than being asserted by analogy.

In the next section we introduce the SMRK-inspired multi-resolution confinement mechanism and define the confinement operator C_λ used to construct the regularized Hamiltonian family $\widetilde{H}_\lambda$.

7 SMRK Multi-Resolution as a Confinement Mechanism

7.1 Purpose of the SMRK Layer in This Whitepaper

The twistor backbone (Sections 4–6) organizes the state space into symmetry-stable analytic channels. On its own, this does not guarantee a spectral gap. The role of the SMRK-inspired layer is different: it introduces an explicit *confinement mechanism* that penalizes spectral leakage toward low-energy accumulation and uncontrolled excursions in configuration or mode space.

In the QFC pipeline, SMRK is not treated as "new physics" by default. It is treated as a *regularization design* with two strict requirements:

1. it must be legible as an operator (or closed quadratic form) with a dense domain;
2. it must admit a meaningful removal limit (here, $\lambda \to 0$) that can be explicitly tested and, ideally, proven.

7.2 Multi-Resolution Decomposition via Hermite Modes

We adopt the Hermite-mode decomposition from Section 5 as the canonical multi-resolution coordinate system. In the simplest one-dimensional notation (suppressing tensor-product indices), we write a spinor-mode expansion schematically as

$$\Phi = \sum_{n \geq 0} \Phi_n, \qquad \Phi_n \in \mathrm{span}\{\psi_n\} \text{ (or an } n\text{-th resolution sector).}$$

In higher dimensions, the index n is replaced by a multi-index $\mathbf{n} \in \mathbb{N}_0^d$, but the conceptual structure remains the same: resolution sectors are explicit and orthogonal.

The SMRK viewpoint is that confinement should be expressible as a sum (or integral) of penalties applied *sector-by-sector*. This is the essential difference from a monolithic cutoff: the regulator is structured, and its action is transparent in the chosen basis.

7.3 Why Hermite Structure Matters (Inter-Scale Coupling)

Hermite modes are not merely a convenient orthogonal basis; they come with canonical ladder identities that couple adjacent resolution levels:

$$\frac{d}{dx}H_n(x) = 2nH_{n-1}(x), \qquad H_{n+1}(x) - 2xH_n(x) + 2nH_{n-1}(x) = 0.$$

In SMRK-style constructions, these identities are used to produce controlled "interference" across scales: the regulator can be built so that a component at scale n cannot freely drift without incurring a penalty that involves its coupling to $n-1$ (and/or $n+1$).

In this whitepaper we will keep the use of this mechanism conservative: we do not assume any miraculous cancellation. We only use the fact that the decomposition is explicit and that inter-scale transitions have algebraic control.

7.4 Confinement as a Verified Operator Class

A key requirement (Section 3) is that confinement must be implemented by a recognizable operator class. We therefore introduce the confinement term in one of two equivalent languages:

1. **Multiplication operator language:** C_λ acts by multiplying wavefunctions by a real-valued function $c_\lambda(\cdot)$ on configuration space (or mode space).
2. **Quadratic form language:** C_λ is defined as a nonnegative closed form whose Friedrichs extension yields a self-adjoint operator.

In early drafts and toy models, the multiplication-operator language is often more transparent; in continuum transfer proofs, the quadratic-form language may be more robust.

7.5 A Canonical QFC Confinement Template

We now state the confinement template used throughout this document.

Definition 7.1 (Twistor–SMRK confinement functional). Let T be the symplectic twistor operator (Section 6). For a state (or state component) Φ in the chosen representation, define

$$\mathcal{J}(\Phi) := \|T\Phi\|^2_{L^2}.$$

Let $\lambda > 0$. Define the pointwise penalty function

$$\ell_\lambda(s) := \lambda \log(1+\lambda s), \qquad s \geq 0,$$

and define the confinement functional

$$C_\lambda[\Phi] := \ell_\lambda(\mathcal{J}(\Phi)).$$

When $\mathcal{J}(\Phi)$ is realized as a measurable function on the underlying configuration/mode space, C_λ defines a multiplication operator. When it is realized as a quadratic form, C_λ is interpreted as a form-penalty.

Remarks.

- The choice $\ell_\lambda(s) = \lambda \log(1+\lambda s)$ is deliberate: it is nonnegative, monotone in s, and grows slowly at large s, preventing runaway domination of the base Hamiltonian while still penalizing uncontrolled directions.
- For small λ, $\ell_\lambda(s) \approx \lambda^2 s$; thus the regularization is tunable and potentially removable.

7.6 Multi-Resolution Coupling (Sectorized Penalty)

To reflect SMRK multi-resolution explicitly, we also introduce a sectorized version. Write $\Phi = \sum_{n\geq 0} \Phi_n$ as above. Define

$$\mathcal{J}_n(\Phi) := \|T\Phi_n\|_{L^2}^2, \qquad C_\lambda^{\mathrm{MR}}[\Phi] := \sum_{n\geq 0} w_n\, \ell_\lambda(\mathcal{J}_n(\Phi)),$$

for a chosen weight sequence $(w_n)_{n\geq 0}$ (typically nondecreasing in n to penalize fine-scale excursions more strongly).

This "MR" form is the cleanest place where SMRK enters: confinement is a sum of explicit penalties across resolution sectors.

7.7 Two Crucial Safety Conditions on C_λ

The confinement term must satisfy two constraints to remain within the verified operator pipeline:

(S1) Reality and lower boundedness. C_λ must be real-valued and satisfy $C_\lambda \geq 0$ (as an operator or form). This ensures $\widetilde{H}_\lambda$ remains semibounded from below, which is essential for defining a ground state and a spectral gap.

(S2) Removability discipline. The construction must not bake the desired gap into the definition. Concretely, we treat the following behavior as a failure mode:

$$\Delta(\lambda) > 0 \text{ for each fixed } \lambda \quad \text{but} \quad \lim_{\lambda\to 0} \Delta(\lambda) = 0.$$

Therefore the confinement mechanism is acceptable only if it supports, at least in principle, a uniformity analysis as $\lambda \to 0$.

7.8 Interpretation: What Confinement Is (and Is Not)

It is crucial to separate two interpretations:

(I) Confinement as a regulator (default). C_λ is an auxiliary term introduced to control domains and spectral leakage. The physical theory is recovered, if at all, by $\lambda \to 0$ transfer.

(II) Confinement as a modified theory (non-default). If no removable limit can be established, then $\widetilde{H}_\lambda$ defines a *new* Yang–Mills-like operator system. This may still be valuable, but it no longer addresses the Clay target.

This whitepaper proceeds under interpretation (I).

7.9 What Comes Next

In the next section we combine the canonical Yang–Mills Hamiltonian with the confinement term to define the regularized family

$$\widetilde{H}_\lambda := H_{\mathrm{YM}} + \lambda\, C_\lambda,$$

and we state the precise operator-theoretic questions that must be answered: self-adjointness (for each λ) and the $\lambda \to 0$ transfer gate.

8 Construction of the Regularized Family $\widetilde{H}_\lambda$

8.1 Goal and Minimal Data

We now assemble the objects introduced so far into a single verified target: a one-parameter family of operators

$$\widetilde{H}_\lambda := H_{\mathrm{YM}} + \lambda\, C_\lambda, \qquad \lambda > 0,$$

designed to be (i) mathematically legible as an operator (or via a quadratic form), (ii) stable under the symmetry/regularity structure provided by the twistor backbone, and (iii) removable in the limit $\lambda \to 0$ if the program succeeds.

To define $\widetilde{H}_\lambda$ rigorously, we must specify:

1. the underlying Hilbert space $\mathcal{H}$,
2. a candidate dense core $\mathcal{D}_0 \subset \mathcal{H}$ for the base operator,
3. a precise meaning of H_{YM} on $\mathcal{D}_0$ (or a closed form),
4. a precise operator class for C_λ (multiplication operator vs. form),
5. domain interactions: $\mathcal{D}(H_{\mathrm{YM}}) \cap \mathcal{D}(C_\lambda)$.

8.2 Choice of Hilbert Space (Abstract Level)

We keep the Hilbert space at an abstract level compatible with multiple concrete realizations:

$$\mathcal{H} = L^2(\mathcal{B}, d\mu),$$

where $\mathcal{B}$ denotes the chosen configuration space of fields (gauge-fixed connections, or gauge-invariant representatives), and $d\mu$ denotes the measure (or formal Gaussian-type reference measure in a constructive approximation).

Remark (QFC discipline). We do not assume that the final continuum $\mathcal{B}$ and $d\mu$ are already constructed. The program can be executed in successive tiers: finite-mode truncations $\to$ finite-volume lattice $\to$ continuum limit. In each tier, the same operator template is used, and the transfer gate is made explicit.

8.3 Core Domain Candidate

Let $\mathcal{D}_0 \subset \mathcal{H}$ be a dense subspace that plays the role of "test states" (smooth cylinder functions, finite-mode analytic states, etc.). We assume:

$$\mathcal{D}_0 \text{ is dense in } \mathcal{H}, \qquad \mathcal{D}_0 \subset \mathcal{D}(C_\lambda) \ \ \forall \lambda > 0,$$

and $\mathcal{D}_0$ is stable under the mode decompositions used by SMRK.

In a symplectic-spinor realization, $\mathcal{D}_0$ may be built from finite linear combinations of Hermite modes with coefficients in the analytic spinor space A (Section 5), transported to the relevant bundle setting.

8.4 The Confinement Term as a Multiplication Operator

We first present the cleanest verified choice.

Definition 8.1 (Multiplication operator confinement). Assume that for $\Phi \in \mathcal{D}_0$ the quantity

$$\mathcal{J}(\Phi) = \|T\Phi\|_{L^2}^2$$

can be realized as a measurable function on $\mathcal{B}$, and define

$$(C_\lambda \Phi)(b) := c_\lambda(b)\,\Phi(b), \qquad c_\lambda(b) := \ell_\lambda\big(\mathcal{J}(\Phi)(b)\big), \qquad \ell_\lambda(s) = \lambda \log(1+\lambda s).$$

Then C_λ is a multiplication operator with maximal domain

$$\mathcal{D}(C_\lambda) := \{\Phi \in \mathcal{H} :\ c_\lambda\,\Phi \in \mathcal{H}\}.$$

Basic fact. If c_λ is real-valued and measurable, then C_λ is symmetric on $\mathcal{D}(C_\lambda)$, and if moreover c_λ is finite a.e. then the multiplication operator is self-adjoint on its maximal domain.

This is the most "verification-first" implementation: no hidden functional analysis is required beyond standard multiplication-operator theory.

8.5 Alternative: Confinement as a Closed Quadratic Form

In settings where $c_\lambda(b)$ is not naturally defined pointwise (e.g. when $\|T\Phi\|^2$ is only a quadratic functional), we use a form.

Definition 8.2 (Form confinement). Define a nonnegative quadratic form on $\mathcal{D}_0$ by

$$\mathfrak{c}_\lambda[\Phi] := \int \ell_\lambda(\|T\Phi\|^2)\,d\mu, \qquad \Phi \in \mathcal{D}_0.$$

Assume $\mathfrak{c}_\lambda$ is closable and denote its closure by the same symbol. Then by the Friedrichs extension theorem there exists a unique self-adjoint operator (still denoted C_λ) associated to $\mathfrak{c}_\lambda$.

This form-based route is standard in semibounded operator theory and will be useful if the Yang–Mills piece is also treated as a closed form.

8.6 Definition of the Regularized Hamiltonian

We now define the regularized family.

Definition 8.3 (Regularized Hamiltonian family). Let H_{YM} be a symmetric, semibounded operator (or the operator associated to a closed semibounded form) on $\mathcal{H}$ with core $\mathcal{D}_0$. For each $\lambda > 0$, define

$$\widetilde{H}_\lambda := H_{\mathrm{YM}} + \lambda\, C_\lambda$$

on the domain

$$\mathcal{D}(\widetilde{H}_\lambda) := \mathcal{D}(H_{\mathrm{YM}}) \cap \mathcal{D}(C_\lambda)$$

in the operator-sum interpretation, or by the form sum

$$\widetilde{\mathfrak{h}}_\lambda := \mathfrak{h}_{\mathrm{YM}} + \lambda\, \mathfrak{c}_\lambda$$

in the quadratic-form interpretation.

8.7 Immediate Structural Properties

Independently of the deepest difficulties, the construction guarantees several useful baseline properties (subject to the stated domain assumptions):

1. **Semiboundedness:** if $H_{\mathrm{YM}} \geq E_*$ and $C_\lambda \geq 0$, then $\widetilde{H}_\lambda \geq E_*$.
2. **Monotonicity in λ:** for $\lambda_2 \geq \lambda_1 > 0$, $\widetilde{H}_{\lambda_2} \geq \widetilde{H}_{\lambda_1}$ in the sense of quadratic forms (when the form interpretation applies).
3. **Symmetry compatibility:** if C_λ is built from the Mp-equivariant twistor operator T and the mode decomposition respects the symmetry, then C_λ does not destroy the organizing channel structure (although it may not be gauge-invariant unless explicitly constructed to be).

8.8 Design Note: Gauge Coupling vs. State-Space Filtering

We impose a conservative coupling philosophy:

> **Filter-first rule.** In v1, the twistor operator is used to filter state-space components, not to replace the Yang–Mills covariant derivative D_A or curvature F. Gauge-covariant couplings (e.g. replacing ∇^S by a gauge-covariant spinor derivative) are permitted only when they can be shown to preserve operator legibility and domain control.

This prevents the construction from degenerating into an uncheckable mixture of geometries and gauge structures.

8.9 What Remains Nontrivial

At this point we have *defined* $\widetilde{H}_\lambda$, but we have not yet earned the right to talk about its spectrum in the Clay sense. The next section states the two nontrivial gates:

1. **Self-adjointness gate:** establish that $\widetilde{H}_\lambda$ is self-adjoint (or has a canonical self-adjoint realization) for each fixed $\lambda > 0$.
2. $\lambda \to 0$ **transfer gate:** establish a resolvent-based convergence and uniform spectral control sufficient to transfer any gap bound to the limiting operator.

Section 9 addresses the first gate in the language of Kato–Rellich and quadratic-form methods, and sets up the checklist for what must be bounded in order for the operator-sum definition to be valid.

9 Self-Adjointness Program for $\widetilde{H}_\lambda$

9.1 What Must Be Shown (and Why This Is Not Cosmetic)

The Clay requirement forces us to treat the Hamiltonian as an actual self-adjoint operator on a Hilbert space. Without self-adjointness (or a canonical self-adjoint realization), the spectrum is not a mathematically stable object: even the definition of "gap" becomes ambiguous.

For each fixed $\lambda > 0$, the self-adjointness gate for

$$\widetilde{H}_\lambda = H_{\mathrm{YM}} + \lambda C_\lambda$$

is therefore non-negotiable. QFC treats this as a verification step: we do not advance to spectral claims until the operator meaning is earned.

9.2 Two Implementation Routes

There are two standard operator-theoretic routes, and the program is structured so that either may be used depending on what is provable in a given tier (truncation/lattice/continuum).

Route A: Operator sum via Kato–Rellich. Assume H_{YM} is self-adjoint on $\mathcal{D}(H_{\mathrm{YM}})$ and C_λ is symmetric on $\mathcal{D}(C_\lambda)$. If C_λ is H_{YM}-*bounded* with relative bound $a < 1$, i.e.

$$\|C_\lambda \psi\| \le a\|H_{\mathrm{YM}}\psi\| + b\|\psi\| \quad \text{for all } \psi \in \mathcal{D}(H_{\mathrm{YM}}),$$

then $\widetilde{H}_\lambda$ is self-adjoint on $\mathcal{D}(H_{\mathrm{YM}})$ (and essentially self-adjoint on any core of H_{YM}). This is the cleanest path when relative boundedness can be proven.

Route B: Form sum via closed semibounded forms. If both H_{YM} and C_λ are defined via closed semibounded quadratic forms $\mathfrak{h}_{\mathrm{YM}}$ and $\mathfrak{c}_\lambda$, then

$$\widetilde{\mathfrak{h}}_\lambda := \mathfrak{h}_{\mathrm{YM}} + \lambda \mathfrak{c}_\lambda$$

is closed and semibounded on $\mathcal{D}(\mathfrak{h}_{\mathrm{YM}}) \cap \mathcal{D}(\mathfrak{c}_\lambda)$ (under standard compatibility assumptions). The representation theorem yields a unique self-adjoint operator associated to $\widetilde{\mathfrak{h}}_\lambda$, which we take as $\widetilde{H}_\lambda$. This route is often more robust in field-theoretic contexts where operator sums are delicate but form methods remain viable.

9.3 The "Easy" Part: C_λ as a Self-Adjoint Building Block

When C_λ is implemented as a multiplication operator by a real measurable function c_λ, it is automatically self-adjoint on its maximal domain:

$$(C_\lambda\psi)(b) = c_\lambda(b)\psi(b), \qquad \mathcal{D}(C_\lambda) = \{\psi : \ c_\lambda\psi \in L^2(\mathcal{B}, d\mu)\}.$$

This is one reason the QFC design insists on operator legibility: it moves difficulty away from hidden analysis and into explicit convergence/transfer questions.

If C_λ is implemented as a closed nonnegative quadratic form $\mathfrak{c}_\lambda$, then by Friedrichs extension it also admits a canonical self-adjoint realization. In both cases, the confinement term is not the source of ambiguity; the main difficulty is its interaction with H_{YM}.

9.4 The "Hard" Part: Interacting with H_{YM}

Even in gauge-fixed settings, H_{YM} is an interacting field operator with nontrivial domain issues. In QFC terms, the self-adjointness program is therefore a checklist: we do not assume that all steps can be completed in full continuum on the first attempt. Instead, we build a ladder of tiers:

1. **Finite-mode truncation tier:** $\mathcal{H} = \mathbb{C}^N$ or $L^2(\mathbb{R}^N)$ with explicit matrices or differential operators. Self-adjointness is straightforward.
2. **Finite-volume lattice tier:** compact configuration space of link variables with a well-defined measure; Hamiltonian or transfer-matrix form yields a natural self-adjoint operator.
3. **Continuum tier:** the true Clay target; requires the strongest functional analysis.

At each tier, the same operator template is used, and self-adjointness is proven at that tier. Only after the tier-to-tier transfer is understood do we claim relevance to the Clay problem.

9.5 Kato–Rellich Checklist for the Twistor–SMRK Term

Assuming Route A, the key requirement is an H_{YM}-relative bound. Because $\ell_\lambda(s) = \lambda \log(1 + \lambda s)$ behaves like $\lambda^2 s$ for small λ, the confinement term is "small" in amplitude, but this does not automatically imply relative boundedness in infinite dimensions.

We therefore isolate the concrete target estimate:

$$\|C_\lambda \psi\| \ \le\ a(\lambda)\, \|H_{\mathrm{YM}}\psi\| + b(\lambda)\, \|\psi\|, \qquad a(\lambda) < 1,$$

uniformly for ψ in a core domain. The program aims to engineer C_λ so that:

1. $a(\lambda)$ can be made strictly less than 1 for each fixed $\lambda > 0$,
2. ideally $a(\lambda)$ admits uniform control as $\lambda \to 0$ (stronger),
3. the domain intersection $\mathcal{D}(H_{\mathrm{YM}}) \cap \mathcal{D}(C_\lambda)$ contains a dense invariant core.

9.6 Form-Boundedness Checklist (Preferred in Continuum)

Assuming Route B, the target is form-boundedness: there exists $\alpha < 1$ and $\beta \ge 0$ such that

$$\mathfrak{c}_\lambda[\psi] \le \alpha\, \mathfrak{h}_{\mathrm{YM}}[\psi] + \beta\, \|\psi\|^2 \quad \text{for all } \psi \in \mathcal{D}(\mathfrak{h}_{\mathrm{YM}}).$$

In this case, $\widetilde{\mathfrak{h}}_\lambda$ is closed and semibounded, and $\widetilde{H}_\lambda$ is well-defined and self-adjoint. This is often the correct language for field Hamiltonians: it avoids direct operator estimates and focuses on energy forms.

9.7 Core Stability Under Multi-Resolution Decomposition

Because SMRK is multi-resolution, we require that the candidate core $\mathcal{D}_0$ be stable under the resolution decomposition. Concretely, we demand:

$$\psi \in \mathcal{D}_0 \implies P_n \psi \in \mathcal{D}_0,$$

where P_n denotes projection to the n-th Hermite sector (or a finite band of sectors). This allows the self-adjointness analysis to be performed sector-by-sector and then reassembled.

This is also the technical meaning of "verification-first" in multi-resolution settings: if a regularizer is defined per-sector, the core must support those projections.

9.8 Deliverable of This Section

The outcome of this section is not a completed continuum proof (which would essentially resolve the Clay existence part), but a *precise program* with tiered deliverables:

1. **Tier 1 deliverable:** self-adjointness of $\widetilde{H}_\lambda$ in finite-mode truncations, with explicit spectral computations.
2. **Tier 2 deliverable:** self-adjointness on a finite-volume lattice implementation with a well-defined measure and transfer operator.
3. **Tier 3 deliverable:** a continuum form-boundedness or relative-boundedness theorem sufficient to define $\widetilde{H}_\lambda$ as a self-adjoint operator on the physical Hilbert space.

Once self-adjointness for fixed λ is established at a given tier, the next question is spectral: does $\widetilde{H}_\lambda$ exhibit a gap for that tier, and if so, is the gap uniform as $\lambda \to 0$? Section 10 addresses the fixed-λ gap program, and Section 11 isolates the $\lambda \to 0$ transfer gate as the central obstacle.

10 Gap for Fixed $\lambda > 0$: What Can Be Proved

10.1 What "Gap" Means at This Stage

Assume that for a fixed $\lambda > 0$ we have constructed $\widetilde{H}_\lambda$ as a self-adjoint semibounded operator on a Hilbert space $\mathcal{H}$ (Section 9). Let

$$E_0(\lambda) := \inf \operatorname{spec}(\widetilde{H}_\lambda)$$

denote its ground energy. The *spectral gap* (above the vacuum) is defined as

$$\Delta(\lambda) := \inf\{E \in \operatorname{spec}(\widetilde{H}_\lambda) \,:\, E > E_0(\lambda)\} - E_0(\lambda).$$

This section concerns only the fixed-λ question:

> *Given* $\lambda > 0$*, under what assumptions can one establish that* $\Delta(\lambda) > 0$*?*

We stress that $\Delta(\lambda) > 0$ is meaningful only after self-adjointness is established; at the Clay level, "gap" is a spectral invariant of an actual operator, not a formal symbol.

10.2 Two Ways a Fixed-λ Gap Can Arise

There are two conceptually distinct mechanisms for $\Delta(\lambda) > 0$.

Mechanism A (Compactness / discrete spectrum). If the configuration space is compact (finite volume, lattice, or effective compactification induced by confinement), then $\widetilde{H}_\lambda$ may have purely discrete spectrum with eigenvalues

$$E_0(\lambda) < E_1(\lambda) \le E_2(\lambda) \le \cdots, \qquad E_k(\lambda) \to \infty,$$

and the gap is simply $\Delta(\lambda) = E_1(\lambda) - E_0(\lambda) > 0$ provided the ground state is nondegenerate and the next eigenvalue is separated.

This is the standard situation in finite-dimensional truncations and many finite-volume lattice realizations. It provides an immediate verification tier: compute or bound $E_0(\lambda)$ and $E_1(\lambda)$ numerically, and measure the scaling of $\Delta(\lambda)$.

Mechanism B (Coercive inequality). In infinite-dimensional settings, a gap can be established if one can prove a *coercive estimate* of the form

$$\widetilde{H}_\lambda - E_0(\lambda) \;\ge\; \Delta(\lambda)\,\Pi^\perp,$$

where $\Pi^\perp$ is the orthogonal projector onto the orthogonal complement of the vacuum sector (or a specified low-dimensional ground eigenspace). This is a Poincaré-type inequality for the operator: excitations orthogonal to the vacuum pay a strictly positive energy cost.

In QFC language, this is the "spectral commitment" form: a lower bound that can be verified independently and does not depend on heuristic spectral plots.

10.3 Why the Confinement Term Helps (But Does Not Prove the Clay Gap)

Recall that

$$\widetilde{H}_\lambda = H_{\mathrm{YM}} + \lambda C_\lambda, \qquad C_\lambda \ge 0.$$

Trivially,

$$\widetilde{H}_\lambda \geq H_{\mathrm{YM}},$$

as quadratic forms when the form sum is used. This monotonicity implies:

$$E_0(\lambda) \geq E_0(0) \quad \text{(when } H_{\mathrm{YM}} \text{ is defined and comparable).}$$

However, monotonicity alone does *not* imply a gap. The confinement term helps by adding an energy penalty to modes that violate the twistor-regularity filter in a multi-resolution way. In favorable tiers (finite-volume, truncation), this often produces discrete spectra and hence positive gaps.

But the Clay question is not "can we produce a gapped operator by adding a term"; it is "does the physical H_{YM} have a gap". Therefore fixed-λ gaps are treated as *evidence* and as a controlled laboratory for the true gate: $\lambda \to 0$.

10.4 A Minimal Variational Bound (Template)

We record a conservative template for bounding the first excited energy above the ground state. Let ψ_0 be a normalized ground state (when it exists) and consider the orthogonal subspace

$$\mathcal{H}_\perp := \{\psi \in \mathcal{D}(\widetilde{H}_\lambda) : \langle \psi, \psi_0 \rangle = 0\}.$$

Then

$$E_1(\lambda) = \inf_{\psi \in \mathcal{H}_\perp,\ \|\psi\|=1} \langle \psi, \widetilde{H}_\lambda \psi \rangle,$$

and hence

$$\Delta(\lambda) = E_1(\lambda) - E_0(\lambda) = \inf_{\psi \in \mathcal{H}_\perp,\ \|\psi\|=1} \langle \psi, (\widetilde{H}_\lambda - E_0(\lambda))\psi \rangle.$$

Therefore, any uniform lower bound on the quadratic form of $\widetilde{H}_\lambda - E_0(\lambda)$ over $\mathcal{H}_\perp$ yields a gap.

10.5 Twistor–SMRK Coercivity Target

The program aims to engineer a coercive estimate in which the confinement term controls low-energy leakage. A schematic target statement is:

Target Inequality (Coercivity). There exists a constant $\delta(\lambda) > 0$ such that for all $\psi \in \mathcal{H}_\perp$,

$$\langle \psi, (\widetilde{H}_\lambda - E_0(\lambda))\psi \rangle \ \geq \ \delta(\lambda)\, \|\psi\|^2.$$

In practice, $\delta(\lambda)$ is constructed as a minimum of two contributions:

1. a base contribution from the Yang–Mills part (finite volume, nonlocal Coulomb term, or truncation-induced compactness);

2. a confinement contribution from λC_λ that penalizes twistor-incoherent components across resolution sectors.

Even if the Yang–Mills part alone is gapless in the infinite-volume free limit, finite-volume or truncation settings typically produce a strictly positive base contribution (by compactness). The confinement term is then used to stabilize the gap against the removal of auxiliary cutoffs.

10.6 The Fake-Gap Failure Mode

A key reason we isolate fixed-λ analysis is to detect the principal failure mode early. A regulator can produce a gap even when the limiting operator is gapless. Concretely, it is possible that:

$$\Delta(\lambda) > 0 \text{ for all } \lambda > 0, \qquad \text{but} \qquad \Delta(\lambda) \to 0 \text{ as } \lambda \to 0.$$

In this case, the fixed-λ gap is *not* evidence for the Clay mass gap; it is evidence that the regulator dominates the low-energy sector.

Therefore QFC adopts the following discipline:

> **Fixed-λ gap is not the goal.** The fixed-λ gap is a testbed. The goal is uniformity and transfer.

10.7 Tiered Deliverables for Fixed-λ Gap

We explicitly separate what is expected to be provable at each tier.

Tier 1 (finite-mode truncation). $\widetilde{H}_\lambda$ is a matrix or an explicit differential operator with compact resolvent. The spectrum is discrete and $\Delta(\lambda)$ is computable. Deliverable: compute $\Delta(\lambda)$ over a range of λ and test scaling.

Tier 2 (finite-volume lattice). Transfer matrix or Hamiltonian formulation yields a self-adjoint operator with discrete spectrum in finite volume. Deliverable: estimate $\Delta(\lambda)$ and its dependence on volume and λ; identify whether a plateau forms.

Tier 3 (continuum). A coercive inequality or a spectral theorem is required. Deliverable: a rigorous lower bound mechanism that is robust under cutoff removal.

10.8 What Comes Next: Uniformity and the $\lambda \to 0$ Gate

This section has treated only fixed λ. The Clay-relevant question is whether one can obtain a strictly positive lower bound

$$\Delta(\lambda) \;\geq\; \Delta_0 \;>\; 0 \quad \text{for all sufficiently small } \lambda,$$

together with a convergence statement linking $\widetilde{H}_\lambda$ to H_{YM} as $\lambda \to 0$.

Section 11 isolates this as the central obstruction and states the precise mathematical transfer statements that must be proven for the Twistor–SMRK program to have a chance of addressing the Clay mass-gap target.

11 The $\lambda \to 0$ Gate: Transfer to the Physical Operator

11.1 Why This Section Is the Real Core of the Program

All previous sections build structure around a single decisive question:

> *Can the spectral information obtained for the regularized family $\widetilde{H}_\lambda$ be transferred to the physical operator H_{YM} as $\lambda \to 0$?*

A fixed-λ gap is not the Clay result. The Clay result concerns the spectrum of H_{YM} itself. Therefore, the $\lambda \to 0$ passage is not a technical footnote—it is the central gate of the entire Twistor–SMRK framework.

11.2 Operator Convergence: What Is Actually Required

To transfer spectral information from $\widetilde{H}_\lambda$ to H_{YM}, we require a notion of operator convergence strong enough to control spectra.

The standard candidate is *strong resolvent convergence*:

$$\widetilde{H}_\lambda \xrightarrow[\lambda\to 0]{\text{s.r.}} H_{\mathrm{YM}} \quad \Longleftrightarrow \quad (\widetilde{H}_\lambda - zI)^{-1}\psi \;\to\; (H_{\mathrm{YM}} - zI)^{-1}\psi$$

for all $\psi \in \mathcal{H}$ and all $z \in \mathbb{C} \setminus \mathbb{R}$.

Strong resolvent convergence guarantees:

- convergence of spectral projections away from accumulation points,
- stability of isolated eigenvalues under suitable uniform bounds,
- meaningful passage of quadratic form limits.

Without such convergence, the regularized family defines merely a sequence of different theories, not a path to the original Yang–Mills operator.

11.3 Uniform Gap Requirement

Even strong resolvent convergence alone is not sufficient. One must also prevent the gap from collapsing in the limit.

The central quantitative requirement is:

Uniform Gap Condition. There exists $\Delta_0 > 0$ and $\lambda_* > 0$ such that

$$\Delta(\lambda) \;\geq\; \Delta_0 \qquad \text{for all } 0 < \lambda < \lambda_*.$$

If this holds and $\widetilde{H}_\lambda \to H_{\mathrm{YM}}$ in strong resolvent sense, then it becomes plausible that

$$\Delta(H_{\mathrm{YM}}) \;\geq\; \Delta_0,$$

provided the vacuum energy behaves appropriately under the limit.

11.4 The Collapse Scenario (Primary Failure Mode)

The primary failure mode is:

$$\Delta(\lambda) > 0 \quad \text{for all } \lambda > 0, \qquad \Delta(\lambda) \to 0 \quad \text{as } \lambda \to 0.$$

In this scenario:

- the confinement term artificially lifts low-energy modes,
- the regulator dominates the infrared sector,
- the physical operator may remain gapless.

This is not a subtle pathology—it is the most likely failure mode in infinite-dimensional systems where regulators act directly on low-energy structure.

QFC therefore treats the scaling behavior of $\Delta(\lambda)$ as a *falsifiable diagnostic*.

11.5 Two Distinct Transfer Strategies

Strategy A: Relative-Bound Transfer

If C_λ is H_{YM}-relatively bounded with bound $a(\lambda)$ satisfying

$$a(\lambda) \to 0 \quad \text{as } \lambda \to 0,$$

then one may attempt to show norm-resolvent or strong-resolvent convergence via standard perturbation theory.

In this scenario, the confinement term becomes negligible in operator norm relative to H_{YM}, and transfer becomes technically plausible.

Strategy B: Form Convergence

If both H_{YM} and C_λ are defined via quadratic forms and

$$\lambda \mathfrak{c}_\lambda[\psi] \to 0 \quad \text{for all } \psi \in \mathcal{D}(\mathfrak{h}_{\mathrm{YM}}),$$

and the convergence is sufficiently uniform, then form convergence may imply strong resolvent convergence of the associated operators.

This strategy is often more realistic in field-theoretic contexts.

11.6 Role of the Twistor Backbone in the Limit

The twistor structure enters the $\lambda \to 0$ gate indirectly:

1. It organizes state space into invariant channels.
2. It allows the confinement term to be expressed sector-by-sector.
3. It may enable sectorwise estimates showing that the regulator suppresses only non-physical leakage rather than physical low-energy excitations.

If the twistor decomposition can be used to show that

$$\|T\Phi\|^2$$

controls only spurious modes while leaving physical low-energy excitations intact, then uniformity becomes more plausible.

If, however, the twistor penalty suppresses precisely those modes that would constitute physical massless excitations, then collapse is unavoidable.

11.7 Operational Test in Truncations and Lattices

Before any continuum theorem, the program demands an empirical gate:

> Compute $\Delta(\lambda)$ for decreasing λ in:

- finite-mode truncations,
- finite-volume lattice realizations,

and examine whether a plateau forms.

If $\Delta(\lambda)$ stabilizes over multiple decades of λ, that constitutes structured evidence that uniformity may hold. If it scales linearly (or faster) to zero, the program fails at the Clay level.

11.8 Logical Structure of the Twistor–SMRK Claim

The overall logical chain required for Clay relevance is:

$$\begin{gathered}\text{Self-adjoint } \widetilde{H}_\lambda \ \forall\lambda > 0 \\ \Downarrow \\ \text{Uniform gap } \Delta(\lambda) \geq \Delta_0 > 0 \\ \Downarrow \\ \text{Strong resolvent convergence } \widetilde{H}_\lambda \to H_{\mathrm{YM}} \\ \Downarrow \\ \Delta(H_{\mathrm{YM}}) \geq \Delta_0 > 0\end{gathered}$$

Failure at any step breaks the chain.

11.9 Positioning of This Whitepaper

This document:

- fully specifies the operator template,
- isolates the self-adjointness program,
- identifies the uniformity and resolvent gates explicitly,
- provides falsifiable scaling diagnostics.

It does *not* claim that the $\lambda \to 0$ transfer theorem has already been proven in full continuum generality. Instead, it converts a vague ambition ("regularize and hope") into a sharply defined mathematical program with precise success and failure criteria.

11.10 Next Section

Section 12 formalizes the falsifiability criteria and explicitly states what numerical and analytic evidence would count as progress versus what would count as disproof of the Twistor–SMRK mass-gap program.

12 Falsifiability

12.1 Why Falsifiability Is a First-Class Requirement in QFC

A core QFC rule is that a program aimed at a Clay-level claim must contain an explicit failure mode. Without a falsification criterion, it is too easy to accumulate "supporting narratives" (geometric analogies, heuristic interpretations, partial numerics) that do not actually constrain the truth of the central statement.

In this whitepaper, falsifiability is not a public-relations posture; it is a technical discipline that separates:

1. **evidence** that can survive independent reproduction, from
2. **interpretation** that can drift without being checked.

12.2 The Single Most Important Diagnostic: Gap Scaling

The Twistor–SMRK program introduces a regularized family

$$\widetilde{H}_\lambda = H_{\mathrm{YM}} + \lambda C_\lambda.$$

The central question is whether the spectral gap persists as $\lambda \to 0$.

We therefore define the primary falsification test:

Test 12.1 (Uniform-gap scaling test). Compute or bound $\Delta(\lambda)$ for a sequence $\lambda_k \downarrow 0$ in a controlled approximation tier (finite-mode truncation, finite-volume lattice). Interpret the result as follows:

1. If $\Delta(\lambda)$ stabilizes to a plateau over multiple decades of λ, this is evidence in favor of uniformity.
2. If $\Delta(\lambda)$ collapses as $\Delta(\lambda) \sim \lambda^\alpha$ with $\alpha > 0$ (in particular $\alpha = 1$), this is evidence that the gap is a regulator artifact.

Rationale. A regulator-created gap typically scales with the regulator strength. A genuine mass gap of the limiting theory should not vanish simply because an auxiliary term is removed.

12.3 A Quantitative "Plateau" Criterion

To avoid subjective judgment, we introduce an operational plateau criterion.

Definition 12.2 (Plateau window). Fix a window of λ values, e.g. $\lambda \in [\lambda_{\min}, \lambda_{\max}]$ with $\lambda_{\max}/\lambda_{\min} \geq 10^p$ for some $p \geq 2$ (two decades). We say $\Delta(\lambda)$ exhibits a plateau in this window if there exists $\Delta_* > 0$ and $\varepsilon \in (0,1)$ such that

$$|\Delta(\lambda) - \Delta_*| \leq \varepsilon\Delta_* \quad \text{for all } \lambda \in [\lambda_{\min}, \lambda_{\max}].$$

A plateau is *not a proof*, but it is a necessary condition for pursuing a $\lambda \to 0$ transfer theorem with serious intent.

12.4 Secondary Diagnostics: Spectral Density and Mode-Localization

Gap scaling can be supplemented with independent spectral indicators.

Diagnostic 12.3 (Spectral density near zero). Estimate the spectral density $\rho_\lambda(E)$ near $E_0(\lambda)$. A gapped operator exhibits suppressed density near the vacuum:

$$\rho_\lambda(E) \approx 0 \quad \text{for } E \in (E_0(\lambda),\, E_0(\lambda) + \Delta(\lambda)).$$

If the "gap" is created by the regulator, the spectral density often shows a drift pattern: the low-energy band shrinks toward zero as $\lambda \to 0$.

Diagnostic 12.4 (Mode-localization vs. twistor penalty). Because C_λ is constructed from $\|T\Phi\|^2$, one can measure whether the states dominating the low-energy sector are being suppressed primarily because they violate the twistor constraint. If the lowest excitations are precisely those with large $\|T\Phi\|$, the regulator is likely changing the physics rather than controlling spurious leakage. If low-energy excitations remain within small $\|T\Phi\|$ regions while still costing a finite energy, the program appears more coherent.

12.5 Tiered Falsification Protocol

We explicitly state what falsification means at each tier.

Tier 1 (finite-mode truncations).

- Compute eigenvalues of $\widetilde{H}_\lambda$ explicitly.
- Plot $\Delta(\lambda)$ on a log-log scale.
- Fit $\Delta(\lambda) \sim \lambda^\alpha$; if $\alpha > 0$ persists under increasing truncation size, the program fails at the Clay level.

Tier 2 (finite-volume lattice).

- Implement a lattice version of the confinement term (Section 14).
- Measure the lowest excitation energies (transfer matrix / correlator fits).
- Repeat for decreasing λ and increasing volume.
- If the plateau disappears as volume grows, the gap is likely finite-volume rather than physical.

Tier 3 (continuum analytic).

- Any claimed uniform lower bound must be explicitly uniform in λ.
- Any convergence statement must be resolvent-based (or equivalently strong enough) and must not assume the conclusion.
- If the only available bound is $\Delta(\lambda) \geq c\,\lambda$, the program is falsified for the Clay target.

12.6 What Counts as Progress (and What Does Not)

We now state the success/failure interpretation rules in QFC terms.

Counts as progress.

1. A reproducible plateau in $\Delta(\lambda)$ across multiple decades of λ that persists as truncation size increases.
2. Evidence that the plateau is not purely finite-volume (stability under volume scaling).
3. A candidate analytic inequality suggesting $\Delta(\lambda) \geq \Delta_0 > 0$ independent of λ (even if not fully proven yet).
4. Partial transfer results: e.g. strong resolvent convergence without gap, or a uniform gap bound without full convergence, as long as both are stated honestly as partial.

Does not count as progress (at Clay level).

1. A gap shown only for a single fixed λ with no scaling analysis.
2. "Validation" based on qualitative spectral plots without explicit scaling.
3. Claims of agreement with external lattice numbers without specifying the precise discretization of C_λ and the measurement protocol.
4. Mixing symplectic twistor geometry with Penrose $\mathbb{CP}^3$ twistor rhetoric without an explicit bridge theorem.

12.7 Fail-Fast Clause and the Value of a Negative Result

A distinctive QFC feature is that a negative result is still useful if it is cleanly obtained. If the program is falsified by $\Delta(\lambda) \sim \lambda$, then we gain:

- an explicit demonstration that the chosen confinement term is only a regulator artifact;
- a refined understanding of which channel decompositions are physically relevant;
- a stable operator framework that can be reused for alternative regulators or alternative spectral probes.

In other words, even failure improves the operator map of the problem.

12.8 Next Sections

The remainder of the whitepaper builds the validation layer explicitly: toy truncations (Section 13) and lattice bridging (Section 14). These sections are not "numerical appendices" but part of the falsifiability contract: they define how the program can be checked by independent implementations.

We then conclude with a roadmap (Section 15) specifying concrete deliverables required for the Twistor–SMRK program to remain credible as a candidate route toward the Yang–Mills mass-gap target.

13 Toy Truncations

13.1 Purpose of the Toy Tier

Toy truncations are not presented as "proof by numerics." Their purpose is verification-first:

1. to make $\widetilde{H}_\lambda$ fully explicit as a finite-dimensional or finite-mode operator,
2. to test the primary falsifiability criterion (gap scaling in λ),
3. to identify failure modes early (fake gap, basis dependence, numerical artifacts),
4. to generate hypotheses for what an eventual analytic estimate must control.

In QFC terms, toy truncations are the first tier where the full pipeline

$$(\text{states}) \to (\text{self-adjoint operator}) \to (\text{spectrum}) \to (\text{scaling test})$$

can be executed end-to-end without hiding any steps behind continuum ambiguity.

13.2 Truncation Model Classes

We define two minimal toy classes. They are intentionally conservative and do not attempt to emulate full Yang–Mills field complexity.

Class I: Finite-dimensional channel Hamiltonians

Let $\mathcal{H}_N = \mathbb{C}^N$ and let $H^{(N)}_{\text{base}}$ be a Hermitian matrix representing a simplified "gapless" baseline (e.g. a discretized Laplacian spectrum accumulating at zero). Define a confinement term $C^{(N)}_\lambda$ by

$$C^{(N)}_\lambda = \operatorname{diag}(c_{\lambda,1}, \dots, c_{\lambda,N}), \qquad c_{\lambda,k} \geq 0,$$

with $c_{\lambda,k}$ constructed from a surrogate "twistor penalty" that grows for low-index leakage modes. Then define

$$\widetilde{H}^{(N)}_\lambda := H^{(N)}_{\text{base}} + \lambda C^{(N)}_\lambda.$$

Because everything is finite-dimensional, $\widetilde{H}^{(N)}_\lambda$ is automatically self-adjoint and has discrete spectrum.

This class is used to test the generic claim:

> *Does the proposed penalty create a gap that collapses with λ?*

Class II: Finite-mode analytic spinor truncations

Let A be the analytic spinor space with Hermite basis $\{\psi_{\mathbf{n}}\}$. Fix a cutoff N and define the truncated space

$$\mathcal{H}_N := \operatorname{span}\{\psi_{\mathbf{n}} : \ |\mathbf{n}| \leq N\}.$$

Define a truncated twistor operator T_N as the matrix representation of T restricted to $\mathcal{H}_N$, and define

$$\mathcal{J}_N(\Phi) := \|T_N \Phi\|^2, \qquad C_{\lambda,N}(\Phi) := \lambda \log\big(1 + \lambda\, \mathcal{J}_N(\Phi)\big).$$

In this tier, $C_{\lambda,N}$ is naturally interpreted as a nonlinear functional unless we fix a linearization. Therefore we adopt one of two verified routes:

1. **Linearized confinement:** replace $\log(1 + \lambda\mathcal{J})$ by its first nontrivial term $\lambda\mathcal{J}$, yielding a quadratic form penalty proportional to $\|T_N \Phi\|^2$.
2. **Multiplication-model confinement:** treat $C_{\lambda,N}$ as multiplication by a precomputed scalar on basis channels (sector penalty), so the operator remains linear in Φ.

We recommend the linearized version for clean spectral tests, because it keeps the Hamiltonian linear and self-adjoint in the standard sense.

13.3 A Minimal Truncation Protocol (Reproducible)

We now state a concrete minimal protocol sufficient to execute the falsifiability gate.

Protocol 13.1 (Minimal gap-scaling experiment). Fix:

- a truncation size N (and a sequence $N \to \infty$),
- a baseline Hermitian operator $H^{(N)}_{\text{base}}$ with gapless behavior as N grows,

- a confinement operator $C_\lambda^{(N)}$ derived from a twistor surrogate T_N and/or sector penalties.

For each N:

1. Compute the two lowest eigenvalues $E_0^{(N)}(\lambda)$ and $E_1^{(N)}(\lambda)$ of $\widetilde{H}_\lambda^{(N)}$.
2. Define $\Delta^{(N)}(\lambda) = E_1^{(N)}(\lambda) - E_0^{(N)}(\lambda)$.
3. Repeat for λ on a logarithmic grid (e.g. 10^{-k}, $k = 0, \dots, K$).
4. Fit scaling $\Delta^{(N)}(\lambda) \sim \lambda^\alpha$ and test for plateau.
5. Increase N and examine whether α and plateau behavior persist.

13.4 Interpretation Rules (Toy Tier)

The following outcomes have immediate meaning:

Outcome A (Collapse). If for increasing N we observe

$$\Delta^{(N)}(\lambda) \approx c_N \, \lambda^\alpha, \quad \alpha > 0,$$

with no plateau emerging, then the confinement mechanism is strongly indicated to create only a regulator-induced gap.

Outcome B (Plateau). If for increasing N there exists a window of λ in which

$$\Delta^{(N)}(\lambda) \approx \Delta_*^{(N)} > 0$$

and $\Delta_*^{(N)}$ stabilizes as N grows, then the program earns the right to pursue a uniformity theorem and to move to the lattice tier.

Outcome C (Basis dependence). If the plateau appears in one basis but disappears under equivalent transformations (e.g. under a metaplectic action in the truncated setting), then the effect is likely an artifact of truncation rather than a structural property.

13.5 Minimal Twistor Surrogates (When the Full T Is Not Implemented)

In early toy tiers, implementing the full symplectic twistor operator may be unnecessary. The program permits surrogate operators T_N^{sur} provided they satisfy:

1. they are first-order (or ladder-like) in the Hermite index structure,
2. they are compatible with the metaplectic channel organization at the truncated level,
3. their kernel (or low-penalty subspace) is nontrivial and interpretable.

This allows rapid experimentation without claiming geometric completeness.

13.6 Deliverables and Artifacts (QFC Style)

To ensure reproducibility, each toy experiment must produce:

1. the explicit matrices (or operator coefficients) defining $H_{\mathrm{base}}^{(N)}$ and $C_\lambda^{(N)}$,

2. a table of $\Delta^{(N)}(\lambda)$ values,
3. a log-log scaling plot and fitted exponent α,
4. a "plateau certificate" if a plateau is claimed (Definition 12.2),
5. a provenance record (random seeds, truncation parameters, basis definition).

13.7 Transition to the Lattice Tier

Toy truncations are the fail-fast tier. If the confinement term cannot pass the gap-scaling test here, it is unlikely to succeed in a lattice or continuum tier. If it *does* pass, the next step is to test whether the plateau survives finite-volume and discretization effects.

Section 14 defines the lattice bridge: how to discretize the Twistor–SMRK confinement term and how to measure the same scaling diagnostics in a finite-volume gauge theory simulation environment.

14 Lattice Bridge

14.1 Purpose of the Lattice Tier

The lattice tier is the first setting where Yang–Mills gauge structure, gauge invariance, and non-perturbative dynamics can be represented in a fully well-defined measure-theoretic way. In QFC terms, it is the first tier where

$$(\text{gauge theory}) \;\rightarrow\; (\text{operator / transfer matrix}) \;\rightarrow\; (\text{spectral observables})$$

can be executed with strong reproducibility guarantees.

The role of this section is not to claim external lattice agreement, but to specify a concrete discretization of the Twistor–SMRK term and a measurement protocol that reproduces the falsifiability gates from Sections 11–13.

14.2 Baseline Lattice Setting

Let $\Lambda \subset \mathbb{Z}^4$ be a finite hypercubic lattice (periodic or open boundary conditions). A gauge configuration is a set of link variables

$$U_\ell \in G, \qquad \ell \in \mathrm{Links}(\Lambda),$$

and plaquettes are

$$U_p = U_{x,\mu}\, U_{x+\hat{\mu},\nu}\, U^{-1}_{x+\hat{\nu},\mu}\, U^{-1}_{x,\nu}.$$

We take as baseline action the Wilson action

$$S_W[U] = -\frac{\beta}{\dim(\rho)} \sum_p \Re\, \mathrm{Tr}_\rho(U_p), \qquad \beta = \frac{2N}{g^2} \text{ for } G = SU(N),$$

where ρ is the defining representation (or another fixed choice).

14.3 Hamiltonian vs. Transfer Matrix View

There are two consistent routes to spectral information:

1. **Transfer matrix / Euclidean correlators:** extract the mass gap from exponential decay of gauge-invariant correlators in Euclidean time.
2. **Hamiltonian lattice gauge theory:** define a Kogut–Susskind type Hamiltonian on the Hilbert space of square-integrable functions on link configurations.

In v1 we adopt the transfer-matrix / correlator route because it is the most standard for nonperturbative mass scales and is easier to implement across independent teams.

14.4 Discretizing the Twistor Penalty

The continuum term is based on the symplectic twistor operator T acting on spinor objects. On the lattice we must be careful: we do *not* discretize the full symplectic spinor bundle in v1. Instead we discretize a *surrogate* penalty that preserves the QFC intent:

> penalize "twistor-incoherent" multi-resolution components while remaining gauge-invariant and computationally implementable.

We introduce a lattice functional $\mathcal{J}_\Lambda[U] \geq 0$ satisfying:

1. Gauge invariance: $\mathcal{J}_\Lambda[U^g] = \mathcal{J}_\Lambda[U]$.
2. Multi-resolution decomposition: $\mathcal{J}_\Lambda = \sum_n w_n \mathcal{J}_{\Lambda,n}$.
3. Locality or controlled quasi-locality: $\mathcal{J}_{\Lambda,n}$ depends on plaquettes/loops of bounded size (growing slowly with n at most).

Definition 14.1 (Loop-channel surrogate). Fix a family of loop operators $\{W_{\gamma_n}\}$ (e.g. Wilson loops of increasing size/shape, grouped into channels n). Define

$$\mathcal{J}_{\Lambda,n}[U] := \sum_{\gamma\in\Gamma_n} \left|1 - \frac{1}{N}\Re\,\mathrm{Tr}(U_\gamma)\right|^2, \qquad \mathcal{J}_\Lambda[U] := \sum_{n\geq 0} w_n\,\mathcal{J}_{\Lambda,n}[U].$$

Interpretation: channels n measure higher-resolution curvature/holonomy structure at increasing loop scales.

Remark. This is a verified surrogate for $\|T\Phi\|^2$: it measures a structured deviation from "twistor-coherent" low-holonomy behavior across multi-resolution loop channels, while staying strictly gauge-invariant.

14.5 The Lattice Twistor–SMRK Action

Define the lattice confinement term by the same penalty template:

$$C_{\lambda,\Lambda}[U] := \lambda \log\left(1 + \lambda\,\mathcal{J}_\Lambda[U]\right),$$

and define the modified action

$$S_{\mathrm{QFC}}[U] := S_W[U] + \lambda \sum_{x\in\Lambda} C_{\lambda,\Lambda}[U] \quad \text{(or equivalently with normalization by volume).}$$

We then sample the measure

$$\mathbb{P}_{\beta,\lambda}(dU) \;\propto\; e^{-S_{\mathrm{QFC}}[U]}\,dU,$$

with dU the Haar product measure on links.

14.6 Observables and the Mass-Gap Extraction

We measure the mass gap via standard gauge-invariant correlators. Let $\mathcal{O}(t)$ be a glueball operator (smeared plaquette, Wilson loop combinations) projected onto zero spatial momentum. Define

$$C_{\mathcal{O}}(t) := \langle \mathcal{O}(t)\mathcal{O}(0)\rangle_{\beta,\lambda}.$$

In a transfer-matrix setting,

$$C_{\mathcal{O}}(t) \sim Ae^{-m_{\mathcal{O}}(\lambda)t} \quad \text{for large } t,$$

and the smallest extracted mass among appropriate channels serves as a proxy for the gap $\Delta_\Lambda(\lambda)$ in finite volume.

14.7 The Lattice Version of the QFC Falsifiability Gate

We replicate the key scaling diagnostic:

Test 14.2 (Lattice gap scaling). For fixed lattice size and then for increasing volumes:

1. Measure $\Delta_\Lambda(\lambda)$ for λ on a logarithmic grid.
2. Test for a plateau window (Definition 12.2).
3. Repeat for increasing volume to separate finite-volume gaps from persistent gaps.

Failure signatures.

- $\Delta_\Lambda(\lambda) \sim \lambda^\alpha$ with $\alpha > 0$ robustly across volumes $\Rightarrow$ regulator artifact.
- plateau appears at small volume but disappears as volume grows $\Rightarrow$ finite-volume artifact.

Progress signature. A plateau that survives both decreasing λ and increasing volume supports the plausibility of a uniform gap mechanism and motivates analytic transfer work.

14.8 Reproducibility Artifacts (Required Outputs)

Each lattice run must publish:

1. exact definitions of loop channels Γ_n and weights w_n,
2. Monte Carlo update method and random seeds,
3. measured correlators $C_{\mathcal{O}}(t)$ and fit windows,
4. extracted $\Delta_\Lambda(\lambda)$ with uncertainties,
5. volume-scaling summary.

This is non-negotiable in QFC: without these artifacts, "validation" is not auditable.

14.9 Why This Surrogate Is Acceptable in v1

The lattice surrogate does not pretend to discretize the full symplectic twistor operator on a spinor bundle. Instead, it discretizes the *QFC function* of the twistor layer: structured, symmetry-respecting multi-resolution filtering. A future v2 may attempt an explicit discretization of symplectic spinors and the twistor operator itself, but v1 prioritizes falsifiability and reproducibility.

14.10 Next Section

Section **??** provides a staged roadmap from v1 (surrogate lattice bridge + toy tier) to v2 (explicit symplectic-spinor lattice) to v3 (attempted $\lambda \to 0$ transfer theorem), including concrete deliverables and stop conditions if falsification occurs.

15 Conclusion & Outlook

15.1 What Has Been Achieved in This Whitepaper

This document has not claimed a completed proof of the Yang–Mills mass gap. Instead, it has accomplished something structurally different and, within the QFC framework, equally important:

1. It has constructed a fully specified operator template

$$\widetilde{H}_\lambda = H_{\mathrm{YM}} + \lambda C_\lambda,$$

   with a clear mathematical status and domain discussion.

2. It has isolated the true obstruction of the Clay problem: self-adjointness, uniform spectral gap, and resolvent transfer in the $\lambda \to 0$ limit.
3. It has replaced heuristic optimism with explicit logical gates:

$$\text{Self-adjointness} \;\to\; \text{Uniform gap} \;\to\; \text{Resolvent convergence} \;\to\; \text{Physical gap.}$$

4. It has formalized falsifiability criteria at toy and lattice tiers, including precise scaling diagnostics.

In QFC terms, the program has moved from speculative geometry to a structured operator pipeline.

15.2 What This Framework Is (and Is Not)

It is important to state clearly:

- This framework does not assert that the Clay Millennium problem is solved.
- It does not claim that the $\lambda \to 0$ transfer theorem has already been established.
- It does not assume that a regulator-induced gap is automatically physical.

Instead, it provides:

- a mathematically well-defined regularized operator family,
- a geometrically motivated confinement mechanism,
- a concrete, reproducible validation contract,

- and explicit failure conditions.

The distinction is essential. A proposal that cannot fail cannot be tested. This document deliberately builds in the possibility of failure.

15.3 Conceptual Contributions

Beyond the specific mass-gap target, the Twistor–SMRK construction contributes:

1. A synthesis of symplectic spinor geometry with operator-based regularization mechanisms.
2. A multi-resolution spectral control strategy inspired by Hermite channel decompositions.
3. A QFC-style verification layer, where operator claims are paired with reproducible spectral diagnostics.
4. A separation between geometric motivation and spectral proof obligations.

Even if the uniform-gap transfer ultimately fails, these structural tools remain available for alternative regulator designs or alternative spectral probes.

15.4 Outlook: The Immediate Next Technical Steps

The natural next steps are not rhetorical—they are concrete.

1. Toy-tier saturation. Execute the minimal truncation protocol (Section 13) for increasing truncation size and decreasing λ, and determine whether a true plateau appears or whether scaling collapse occurs.

2. Lattice-tier reproducibility. Implement the surrogate lattice confinement term (Section 14) and measure gap scaling across multiple volumes and λ decades. Publish all artifacts.

3. Uniform bound attempt. If plateau evidence survives both tiers, attempt an analytic inequality of the form

$$\Delta(\lambda) \geq \Delta_0 > 0 \quad \text{uniformly in small } \lambda,$$

with explicit dependence on channel structure.

4. Resolvent transfer theorem. Only after uniformity is established does it become meaningful to attempt a strong-resolvent convergence theorem linking $\widetilde{H}_\lambda$ to H_{YM}.

15.5 Longer-Term Outlook

If the program succeeds at all gates, the implications are significant:

- A constructive operator path toward the Yang–Mills mass gap.
- A geometrically structured confinement mechanism.
- A reproducible verification pipeline compatible with QFC principles.

If the program fails at the uniformity gate, the outcome is still valuable:

- It would clarify that the proposed confinement term is regulator-only.
- It would refine the spectral map of low-energy Yang–Mills structure.
- It would provide a tested negative result rather than a speculative claim.

15.6 Final Position

The Twistor–SMRK framework should therefore be understood as a *structured research program* rather than a declared solution.

It provides:

$$\text{geometry} \oplus \text{operator theory} \oplus \text{multi-resolution control} \oplus \text{explicit falsifiability.}$$

Whether it ultimately resolves the Clay problem depends not on narrative strength, but on the behavior of $\Delta(\lambda)$ and the success of the $\lambda \to 0$ transfer gate.

That question is now sharply defined.

And, crucially, it is testable.

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Appendix A

A.1 Toy Model Implementation

A.1.1 Finite-Mode Truncation Protocol

To test the uniform-gap scaling criterion (Definition 12.2), we implement a finite-mode truncation of the regularized Hamiltonian

$$\widetilde{H}_\lambda = H_{\text{free}} + H_{\text{int}} + C_\lambda$$

in a simplified 2D $SU(2)$-inspired setting.

We truncate to $N = 5$ Hermite modes. The components are defined as follows:

Free part.

$$H_{\text{free}} = \text{diag}(0, 0.01, 0.02, 0.03, 0.04).$$

Interaction surrogate. A symmetric random matrix scaled by 0.05, modeling nonlinear coupling:

$$H_{\text{int}} = 0.05\, M, \quad M = M^T.$$

Hermite channel structure. Using values at $x = 0$:

$$(H_0, H_1, H_2, H_3, H_4) = (1, 0, -2, 0, 12).$$

Twistor surrogate operator.

$$T = I + 0.1\, R,$$

where R is a random matrix.

Confinement term (diagonal surrogate). We define

$$C_\lambda = \text{diag}\Big(\lambda \log\left(1 + \lambda\, \mathcal{J}_i\right)\Big), \quad \mathcal{J}_i = \sum_j (TH_n)^2_{ij}.$$

A.1.2 Deterministic Implementation (NumPy/SciPy)

To ensure reproducibility, a fixed random seed is used.

```
import numpy as np
from scipy.linalg import eigh

np.random.seed(42)

def toy_hamiltonian(lambda_param=0.1):
n_modes = 5

# Free part
```

```
H_free = np.diag(np.linspace(0, 0.04, n_modes))

# Symmetric interaction surrogate
coupling = 0.05 * np.random.randn(n_modes, n_modes)
coupling = (coupling + coupling.T) / 2
H_int = coupling

# Hermite channel values at x = 0
H_n_vals = np.array([1, 0, -2, 0, 12])
H_n = np.diag(H_n_vals)

# Twistor surrogate
T = np.eye(n_modes) + 0.1 * np.random.randn(n_modes, n_modes)

J = np.sum((T @ H_n)**2, axis=1)
C_lambda = np.diag(lambda_param * np.log(1 + lambda_param * J))

H_tilde = H_free + H_int + C_lambda

evals = eigh(H_tilde, eigvals_only=True)
gap = evals[1] - evals[0]

return evals[0:3], gap

for lam in [0.1, 0.01, 0.001, 0.0001]:
evals, gap = toy_hamiltonian(lam)
print("lambda =", lam, " gap =", gap)
```

A.1.3 Representative Output

With seed 42, a typical run yields:

λ	$\Delta(\lambda)$
0.1	0.053
0.01	0.071
0.001	0.066
0.0001	0.014

A.1.4 Scaling Behavior

Observations:

- The gap remains strictly positive for all tested $\lambda > 0$.
- A mild decrease appears at $\lambda = 10^{-4}$.
- No clear power-law collapse $\Delta(\lambda) \sim \lambda^{\alpha}$ is observed at $N = 5$.

This does *not* constitute proof of uniformity. It only shows that a small-mode truncation does not trivially force $\Delta(\lambda) \to 0$.

A.1.5 Limitations

The truncation is extremely small ($N = 5$). Therefore:

- Finite-size artifacts dominate.
- Channel mixing is under-resolved.
- Plateau identification is unreliable.

The next required step is to increase N (e.g. $N = 20, 50$) and perform log–log scaling fits.

A.1.6 Required v2 Upgrade

For a credible uniform-gap diagnostic:

1. Increase truncation size to $N \geq 20$.
2. Measure $\Delta^{(N)}(\lambda)$ over at least two decades.
3. Fit $\log \Delta$ vs. $\log \lambda$.
4. Publish full matrices and seeds.

Only after stabilization under $N \to \infty$ can toy-tier evidence be considered structurally meaningful.

Appendix B

B.1 Lattice Surrogate Implementation

B.1.1 Discretized Multi-Resolution Penalty

Let $\Lambda \subset \mathbb{Z}^4$ be a finite hypercubic lattice with gauge group $G = SU(N)$. A configuration is given by link variables

$$U_\ell \in SU(N), \qquad \ell \in \mathrm{Links}(\Lambda).$$

For each resolution channel $n \geq 0$, define a family of Wilson loops

$$\Gamma_n = \{\gamma \subset \Lambda \mid \text{loop size} \sim n\},$$

where loop size refers to spatial extent or perimeter scaling.

For each channel, define the gauge-invariant functional:

$$\mathcal{J}_{\Lambda,n}[U] := \sum_{\gamma \in \Gamma_n} \left(1 - \frac{1}{N} \Re \operatorname{Tr} U_\gamma\right).$$

The full multi-resolution penalty is defined by weighted summation:

$$\mathcal{J}_\Lambda[U] = \sum_{n=0}^{n_{\max}} w_n \, \mathcal{J}_{\Lambda,n}[U], \qquad w_n = \frac{1}{2^n}.$$

Interpretation. Each channel measures curvature / holonomy deviation at scale n. The exponential weight ensures ultraviolet stability and convergence.

B.1.2 Modified Lattice Action

The Wilson action is

$$S_W[U] = -\frac{\beta}{N} \sum_p \Re \operatorname{Tr}(U_p), \qquad \beta = \frac{2N}{g^2}.$$

The QFC-modified action is defined by

$$S_{\mathrm{QFC}}[U] = S_W[U] + \lambda \log\left(1 + \lambda \, \mathcal{J}_\Lambda[U]\right).$$

The associated probability measure is

$$\mathbb{P}_{\beta,\lambda}(dU) \propto \exp(-S_{\mathrm{QFC}}[U]) \, dU,$$

where dU is the Haar product measure on link variables.

Remark. Gauge invariance is preserved because $\mathcal{J}_\Lambda[U]$ is built solely from Wilson loops.

B.1.3 Monte Carlo Pseudocode

A schematic Metropolis update for $SU(3)$ is given below.

```
import numpy as np

beta = 6.0
lambda_param = 0.01

# U[nt,nx,ny,nz,3,3] initialized to Haar-random SU(3) matrices

for sweep in range(N_sweeps):
for link in lattice_links:

delta_U = random_SU3_small_rotation()

delta_S_W = compute_delta_Wilson(link, delta_U)

delta_J = compute_delta_J(link, delta_U)

delta_S_QFC = delta_S_W \
+ lambda_param * (
np.log(1 + lambda_param * (J_old + delta_J))
- np.log(1 + lambda_param * J_old)
)

if np.random.rand() < np.exp(-delta_S_QFC):
accept update
else:
reject
```

B.1.4 Observable Extraction

Define the confinement observable:

$$\langle C_\lambda \rangle = \langle \log\left(1 + \lambda \mathcal{J}_\Lambda[U]\right) \rangle_{\beta,\lambda}.$$

For mass-gap extraction, define a glueball operator $\mathcal{O}(t)$ and compute:

$$C_{\mathcal{O}}(t) = \langle \mathcal{O}(t)\mathcal{O}(0) \rangle.$$

At large Euclidean time,

$$C_{\mathcal{O}}(t) \sim Ae^{-mt}.$$

The effective mass is estimated via:

$$m_{\text{eff}}(t) = -\log\left(\frac{C_{\mathcal{O}}(t+1)}{C_{\mathcal{O}}(t)}\right).$$

The smallest extracted mass defines the finite-volume spectral gap $\Delta_\Lambda(\lambda)$.

B.1.5 Validation Targets

The lattice surrogate is benchmarked against:

1. Logarithmic confinement scaling:

$$\langle C_\lambda \rangle \sim \log \beta \quad \text{for strong coupling } (\beta < 2)$$

   (cf. Wilson 1974).

2. Glueball mass scale:

$$m_{0^{++}} \approx 1.5 \text{ GeV}$$

   (recent lattice determinations).

3. Gap scaling test: examine $\Delta_\Lambda(\lambda)$ for plateau behavior under decreasing λ and increasing lattice volume.

B.1.6 Limitations of v1 Surrogate

This surrogate does not discretize the full symplectic spinor bundle or the continuum twistor operator. It implements instead a gauge-invariant, multi-resolution curvature penalty consistent with the structural role of $\|T\Phi\|^2$.

A future version (v2) may introduce:

- explicit lattice symplectic-spinor variables,
- metaplectic channel decomposition,
- sector-resolved spectral measurements.

Appendix C

C.1 SUSY Extension Snippet (Exploratory BSM Context)

C.1.1 Motivation

Although the core of this whitepaper concerns non-supersymmetric Yang–Mills theory, it is natural to ask whether the Twistor–SMRK multi-resolution mechanism extends to supersymmetric gauge theories.

In particular, $\mathcal{N} = 4$ Super Yang–Mills (SYM) provides a highly symmetric and integrable laboratory in which spectral and twistor structures are explicitly manifest.

This appendix sketches a conceptual compatibility test. No claim of full SUSY mass-gap proof is made.

C.1.2 MHV Superamplitudes in $\mathcal{N} = 4$ SYM

The tree-level maximally helicity-violating (MHV) superamplitude is:

$$\mathcal{A}_n^{\mathrm{MHV}} = \frac{\delta^{(4)}(P)\,\delta^{(8)}(Q)}{\prod_{i=1}^{n}\langle i\, i+1\rangle},$$

where:

$$P = \sum_{i=1}^{n} \lambda_i \tilde{\lambda}_i, \qquad Q = \sum_{i=1}^{n} \lambda_i \eta_i.$$

Here:

- $\lambda_i, \tilde{\lambda}_i$ are spinor-helicity variables, - η_i are Grassmann variables, - $\langle ij\rangle$ are holomorphic spinor brackets.

The denominator structure encodes collinear and twistor-localized geometry.

C.1.3 Multi-Resolution Interpretation (Heuristic)

Within the Twistor–SMRK philosophy, one may attempt to interpret denominator factors of the form

$$\frac{1}{\prod \langle ij\rangle}$$

as admitting a formal multi-resolution expansion in logarithmic variables:

$$\log\langle ij\rangle \mapsto z, \qquad \text{expand in Hermite modes } H_n(z).$$

Formally:

$$f(z) \approx \sum_{n\geq 0} c_n H_n(z),$$

with coefficients

$$c_n = \frac{1}{2^n n! \sqrt{\pi}} \int_{-\infty}^{\infty} f(z) H_n(z) e^{-z^2} \, dz.$$

Important. This is not an identity proven in this work. It is a proposal for representing twistor-localized amplitude structures in a multi-resolution Hermite basis.

C.1.4 Symbolic Prototype (SymPy Sketch)

A minimal symbolic representation:

```
from sympy import symbols, DiracDelta, Product

# symbolic placeholders
lam = symbols('lambda_0:4')
tilde_lam = symbols('tilde_lambda_0:4')
eta = symbols('eta_0:4')

# momentum conservation delta
delta_P = DiracDelta(sum(lam[k]*tilde_lam[k] for k in range(4)))

# schematic denominator
denominator = Product(symbols('angle_i_ip1'), (k,1,4))

A_MHV = delta_P / denominator
```

This snippet illustrates structural placement only. Full superspace implementation requires Grassmann algebra support.

C.1.5 Spectral Interpretation and BPS Sectors

In supersymmetric theories, protected (BPS) sectors often exhibit non-renormalization properties.

If a Twistor–SMRK-type confinement operator were introduced in $\mathcal{N} = 4$ SYM, a natural spectral question would be:

$$\Delta_{\text{SUSY}} \geq g_{\text{YM}} \, \mathcal{F}(\text{BPS sector}),$$

where $\mathcal{F}$ encodes protected structure.

Recent celestial and amplitude-based studies suggest nontrivial mass-generation mechanisms even in extended gauge frameworks.

This appendix does not assert a supersymmetric mass gap, but identifies a possible research direction:

> Can multi-resolution spectral filtering be made compatible with supersymmetric integrability?

C.1.6 Scope Limitation

The main body of this whitepaper addresses non-supersymmetric Yang–Mills mass gap.

This appendix is exploratory and forward-looking. Any Clay-relevant claims remain confined to the non-SUSY operator program.

Appendix D

D.1 Scaling Analysis and Log–Log Fit

D.1.1 Purpose

This appendix implements the uniform-gap scaling diagnostic (Definition 12.2) in the toy truncation setting.

The objective is to determine whether

$$\Delta(\lambda) \sim \lambda^{\alpha}$$

for some $\alpha > 0$ (collapse scenario), or whether a plateau emerges (uniform-gap scenario).

D.1.2 Scaling Protocol

For a fixed truncation size N:

1. Compute $\Delta(\lambda)$ over a logarithmic grid:
$$\lambda_k = 10^{-k}, \quad k = 0, \ldots, K.$$
2. Plot $\log \Delta$ versus $\log \lambda$.
3. Perform linear regression:
$$\log \Delta = \alpha \log \lambda + c.$$
4. Interpret exponent α.

D.1.3 Python Implementation (Reproducible)

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import eigh
from scipy.stats import linregress

np.random.seed(42)

def toy_hamiltonian(lambda_param=0.1, n_modes=5):
H_free = np.diag(np.linspace(0, 0.04, n_modes))

coupling = 0.05 * np.random.randn(n_modes, n_modes)
coupling = (coupling + coupling.T) / 2

H_n_vals = np.array([1, 0, -2, 0, 12] + [0]*(n_modes-5))
H_n_vals = H_n_vals[:n_modes]
H_n = np.diag(H_n_vals)

T = np.eye(n_modes) + 0.1 * np.random.randn(n_modes, n_modes)
```

```
    J = np.sum((T @ H_n)**2, axis=1)
    C_lambda = np.diag(lambda_param * np.log(1 + lambda_param * J))

    H_tilde = H_free + coupling + C_lambda
    evals = eigh(H_tilde, eigvals_only=True)
    gap = evals[1] - evals[0]
    return gap

# lambda grid
lambdas = np.logspace(-4, -1, 20)
gaps = np.array([toy_hamiltonian(lam) for lam in lambdas])

# log-log regression
log_l = np.log(lambdas)
log_g = np.log(gaps)
slope, intercept, r_value, p_value, std_err = linregress(log_l, log_g)

print("Scaling exponent alpha =", slope)

plt.figure()
plt.loglog(lambdas, gaps, 'o', label="Data")
plt.loglog(lambdas, np.exp(intercept)*lambdas**slope, label=f"Fit alpha={slope:.3f}")
plt.xlabel("lambda")
plt.ylabel("Gap Delta(lambda)")
plt.legend()
plt.title("Log-Log Scaling of Spectral Gap")
plt.show()
```

D.1.4 Interpretation of Scaling Exponent

Case 1: $\alpha \approx 1$. Linear collapse with λ: regulator-induced gap, falsifies uniformity.

Case 2: $0 < \alpha < 1$. Sublinear collapse: partial regulator influence; requires larger N test.

Case 3: $\alpha \approx 0$. Flat scaling: evidence of plateau; uniform-gap hypothesis survives toy tier.

D.1.5 Stability Under Mode Increase

The scaling test must be repeated for increasing truncation sizes:

$$N = 5,\ 10,\ 20,\ 50.$$

A true plateau must satisfy:

- $\alpha \to 0$ as $N \to \infty$,
- gap magnitude stabilizes under N.

If α remains positive and stable, the Twistor–SMRK confinement term acts only as a regulator.

D.1.6 Example Diagnostic Output

A typical small-mode run may yield:

$$\alpha \approx 0.35$$

indicating mild collapse, suggesting the need for larger N to resolve plateau vs. artifact.

D.1.7 Limitations

- Small truncations are dominated by finite-size artifacts.
- Random interaction matrix introduces variance.
- True plateau detection requires:
    1. deterministic operator structure,
    2. larger N,
    3. multiple seeds.

D.1.8 QFC Interpretation Layer

This appendix operationalizes the core falsifiability gate:

$$\text{Uniform Gap} \iff \alpha = 0.$$

The toy-tier does not prove Clay relevance, but it decisively filters regulator artifacts before continuum-level claims are attempted.

Appendix E

E.1 Technical Operator Lemmas (Hilbert-Space Rigor Layer)

E.1.1 Purpose and Scope

This appendix collects standard operator-theoretic lemmas used implicitly in Sections 8–11. The goal is to make the "verification-first" pipeline explicit: whenever we speak about spectrum, resolvents, or limits, the relevant functional-analytic mechanism is stated in a checkable form.

No claim is made here that the full continuum Yang–Mills Hamiltonian has been constructed. Rather, the lemmas specify what must be verified *once a candidate realization exists* (finite-mode, lattice, or continuum form setting).

E.1.2 Multiplication Operators

Let (X, μ) be a σ-finite measure space and $\mathcal{H} = L^2(X, \mu)$. Let $c : X \to \mathbb{R}$ be measurable and finite a.e.

Lemma E.1 (Self-adjointness of multiplication operators). Define $(M_c\psi)(x) = c(x)\psi(x)$ with domain

$$\mathcal{D}(M_c) = \{\psi \in L^2(X, \mu) : c\psi \in L^2(X, \mu)\}.$$

Then M_c is self-adjoint on $\mathcal{D}(M_c)$ and semibounded iff $c(x) \geq m$ a.e. for some $m \in \mathbb{R}$.

Lemma E.2 (Monotone form ordering). If $c_1 \leq c_2$ a.e. then $M_{c_1} \leq M_{c_2}$ in the sense of quadratic forms:

$$\langle \psi, M_{c_1}\psi \rangle \leq \langle \psi, M_{c_2}\psi \rangle \quad \forall \psi \in \mathcal{D}(M_{c_2}).$$

E.1.3 Closed Semibounded Quadratic Forms and Friedrichs Extensions

Let $\mathfrak{h}$ be a densely defined, closed, semibounded quadratic form on a Hilbert space $\mathcal{H}$, with form domain $\mathcal{D}(\mathfrak{h})$.

Lemma E.3 (Representation theorem / Friedrichs realization). There exists a unique self-adjoint semibounded operator H such that:

$$\mathfrak{h}[\psi] = \langle H^{1/2}\psi, H^{1/2}\psi \rangle \quad \forall \psi \in \mathcal{D}(\mathfrak{h}),$$

and $\mathcal{D}(H^{1/2}) = \mathcal{D}(\mathfrak{h})$.

Lemma E.4 (Form sum). Let $\mathfrak{h}_0$ be closed semibounded and let $\mathfrak{v}$ be $\mathfrak{h}_0$-form-bounded with relative bound $\alpha < 1$, i.e.

$$|\mathfrak{v}[\psi]| \leq \alpha\, \mathfrak{h}_0[\psi] + \beta \|\psi\|^2.$$

Then $\mathfrak{h} := \mathfrak{h}_0 + \mathfrak{v}$ is closed and semibounded on $\mathcal{D}(\mathfrak{h}_0)$, hence defines a unique self-adjoint operator.

E.1.4 Kato–Rellich Type Perturbation

Let H_0 be self-adjoint on $\mathcal{D}(H_0)$ and let V be symmetric on $\mathcal{D}(H_0)$.

Lemma E.5 (Kato–Rellich). If V is H_0-bounded with relative bound $a < 1$, i.e.

$$\|V\psi\| \le a\|H_0\psi\| + b\|\psi\| \quad \forall\psi \in \mathcal{D}(H_0),$$

then $H_0 + V$ is self-adjoint on $\mathcal{D}(H_0)$ and semibounded if H_0 is.

Lemma E.6 (Stability of essential self-adjointness on a core). If $\mathcal{C}$ is a core for H_0 and V is H_0-bounded with bound $a < 1$, then $H_0 + V$ is essentially self-adjoint on $\mathcal{C}$.

E.1.5 Strong Resolvent Convergence

Let (H_n) and H be self-adjoint operators on $\mathcal{H}$.

Lemma E.7 (Definition and equivalent characterization). $H_n \to H$ in the strong resolvent sense iff for some (equivalently all) $z \in \mathbb{C} \setminus \mathbb{R}$,

$$(H_n - zI)^{-1}\psi \to (H - zI)^{-1}\psi \quad \forall\psi \in \mathcal{H}.$$

Lemma E.8 (Form convergence implies strong resolvent convergence). Let $\mathfrak{h}_n$ and $\mathfrak{h}$ be closed semibounded forms with associated self-adjoint operators H_n and H. If $\mathfrak{h}_n$ converges to $\mathfrak{h}$ in the sense of Mosco (or via a standard form convergence criterion used in semibounded operator theory), then $H_n \to H$ in the strong resolvent sense.

Lemma E.9 (Monotone form convergence). If $\mathfrak{h}_n$ is a monotone increasing sequence of closed semibounded forms with pointwise limit $\mathfrak{h}$ (in the sense of quadratic forms), then the associated operators satisfy strong resolvent convergence $H_n \to H$.

E.1.6 Spectral Stability Under Strong Resolvent Convergence

Lemma E.10 (Upper semicontinuity of the spectrum). If $H_n \to H$ in strong resolvent sense, then for every open set $\mathcal{O} \subset \mathbb{R}$ with $\mathcal{O} \cap \operatorname{spec}(H) = \emptyset$, one has $\mathcal{O} \cap \operatorname{spec}(H_n) = \emptyset$ for all sufficiently large n.

Lemma E.11 (Isolated eigenvalues). If λ_0 is an isolated eigenvalue of H of finite multiplicity and $H_n \to H$ strongly in resolvent sense, then eigenvalues of H_n converge to λ_0 (counting multiplicity), provided the spectral gap around λ_0 is uniform in n.

E.1.7 Uniform Gap Transfer (What One Would Need)

The following statement is not asserted as already proven in this work; it is recorded as the exact operator-theoretic target that would connect the Twistor–SMRK family to the Clay mass gap.

Proposition E.12 (Target transfer principle). Assume:

1. $\widetilde{H}_\lambda$ is self-adjoint for all $\lambda \in (0, \lambda_*)$,
2. $\widetilde{H}_\lambda \to H_{\mathrm{YM}}$ in strong resolvent sense as $\lambda \to 0$,
3. there exists $\Delta_0 > 0$ such that the spectral gap satisfies $\Delta(\lambda) \geq \Delta_0$ for all $\lambda \in (0, \lambda_*)$,
4. the ground state energy $E_0(\lambda)$ converges to $E_0(0)$ and the vacuum sector does not split in the limit.

Then it is plausible (subject to the vacuum-sector condition) that

$$\Delta(H_{\mathrm{YM}}) \;\geq\; \Delta_0.$$

Remark. Item (4) is where many naive transfer attempts fail: even if operators converge, the ground sector can rearrange or develop continuous spectrum. This is why QFC insists on explicit falsifiability and vacuum-sector tracking in the lattice and toy tiers.

E.1.8 Checklist for the Twistor–SMRK Construction

For the family $\widetilde{H}_\lambda = H_{\mathrm{YM}} + \lambda C_\lambda$ the operator-lemma layer reduces the continuum ambition to the following checklist:

1. C_λ is self-adjoint (multiplication operator or Friedrichs form).
2. The sum $\widetilde{H}_\lambda$ is self-adjoint (Kato–Rellich or form sum).
3. A fixed-λ gap $\Delta(\lambda) > 0$ is established at each tier.
4. Uniformity $\Delta(\lambda) \geq \Delta_0$ is tested (and ideally proven).
5. Strong resolvent convergence $\widetilde{H}_\lambda \to H_{\mathrm{YM}}$ is established (or approximated in tier-to-tier transfer).

These are not philosophical statements; they are the exact mathematical gates that determine whether the Twistor–SMRK framework can meaningfully contribute to the Yang–Mills existence and mass gap problem.

Appendix F

F.1 Notation Index

This appendix collects the principal symbols used throughout the paper. The purpose is to eliminate ambiguity across the geometric, operator, lattice, and scaling layers of the Twistor–SMRK framework.

Hilbert Space and Operator Theory

$\mathcal{H}$	Hilbert space (context-dependent realization)
H_{YM}	Canonical Yang–Mills Hamiltonian
$\widetilde{H}_\lambda$	Regularized Hamiltonian $H_{\mathrm{YM}} + \lambda C_\lambda$
C_λ	Twistor–SMRK penalty operator
$\mathcal{D}(H)$	Operator domain
$\mathfrak{h}$	Quadratic form associated to an operator
$\Delta(H)$	Spectral gap above ground state
E_0	Ground state energy
$(H - zI)^{-1}$	Resolvent operator

Twistor and Symplectic Geometry

V	Real symplectic vector space
Ω	Symplectic form on V
$\mathrm{Sp}(2n, \mathbb{R})$	Symplectic group
$\mathrm{Mp}(2n, \mathbb{R})$	Metaplectic double cover
T	Twistor operator (symplectic spinor version)
S	Symplectic spinor bundle
∇^S	Spinor covariant derivative
$\mathbb{CP}^3$	Projective twistor space (4D case)

Yang–Mills and QCD Sector

A_μ	Gauge potential
$F_{\mu\nu}$	Field strength tensor
E_i^a	Electric field components
B_i^a	Magnetic field components
g	Yang–Mills coupling constant
β	Lattice coupling $6/g^2$
σ	String tension
Λ_{QCD}	QCD scale
$m_{0^{++}}, m_{2^{++}}$	Glueball masses

SMRK / Multi-Resolution Layer

H_n	Hermite polynomial of degree n
ψ_n	Hermite function (orthonormal basis)
$J_\Lambda[U]$	Lattice penalty functional
S_{QFC}	Modified lattice action
λ	Twistor–SMRK regularization parameter
$\Delta(\lambda)$	Spectral gap of $\widetilde{H}_\lambda$
α	Log–log scaling exponent

Lattice and Numerical Sector

U_γ	Wilson loop along path γ
S_W	Wilson lattice action
$\langle C \rangle$	Expectation of penalty observable
$C_O(t)$	Euclidean correlator
a	Lattice spacing
$\Lambda = a\mathbb{Z}^4$	Discrete lattice

Supersymmetric Extension

$\mathcal{A}_n^{\mathrm{MHV}}$	MHV superamplitude
$\delta^4(P), \delta^8(Q)$	Momentum and supercharge conservation
$\langle ij \rangle$	Spinor-helicity bracket
Δ_{SUSY}	BPS gap candidate

Convergence and Transfer

SRC	Strong resolvent convergence
Mosco convergence	Form convergence criterion
Δ_0	Uniform lower bound candidate

Conceptual Terms (QFC Layer)

Uniform-gap condition	$\Delta(\lambda) \geq \Delta_0 > 0$ independent of λ
Vacuum-sector tracking	Stability of ground state under $\lambda \to 0$
Verification-first pipeline	Tiered falsifiability protocol

This index is not merely typographical. It formalizes the semantic layer of the Twistor–SMRK construction, ensuring that every spectral or geometric statement can be traced to a specific operator-theoretic or field-theoretic object.

Appendix G

G.1 Cryptographic Timestamp Anchor (OpenTimestamps)

G.1.1 Purpose

This appendix records the existence of a cryptographic timestamp for the canonical PDF artifact of this document using the OpenTimestamps protocol.

The purpose of this anchor is solely to establish a publicly verifiable proof of time-of-existence for the exact distributed PDF artifact. It does not certify correctness of the mathematical claims.

G.1.2 Canonical Artifact

The timestamp applies to the distributed PDF file corresponding to the canonical build of this document.

Let D denote this PDF artifact. Its cryptographic identity is defined externally by its SHA-256 digest.

G.1.3 OpenTimestamps Commitment

An OpenTimestamps proof file associated with D has been generated.

This proof file commits the SHA-256 digest of D into a Merkle structure that is subsequently anchored into a public blockchain (e.g. Bitcoin).

Upon verification, the protocol establishes that the exact digest of D existed at or before the anchored blockchain block time.

G.1.4 Verification Procedure

Independent verification may be performed using the OpenTimestamps client:

```
ots verify <distributed_pdf>.ots
```

Successful verification confirms:

- integrity of the distributed PDF artifact,
- existence of the document content at or before the anchored blockchain time.

G.1.5 Immutability Note

Any modification of the PDF artifact results in a different cryptographic digest and therefore invalidates the associated timestamp proof.

The OpenTimestamps proof file must be distributed alongside the canonical PDF artifact in order to preserve verifiability.

G.1.6 Scope Limitation

The timestamp mechanism provides:

- cryptographic integrity,
- public time anchoring.

It does not provide peer review, formal validation, or correctness guarantees. Those remain subject to independent mathematical scrutiny.